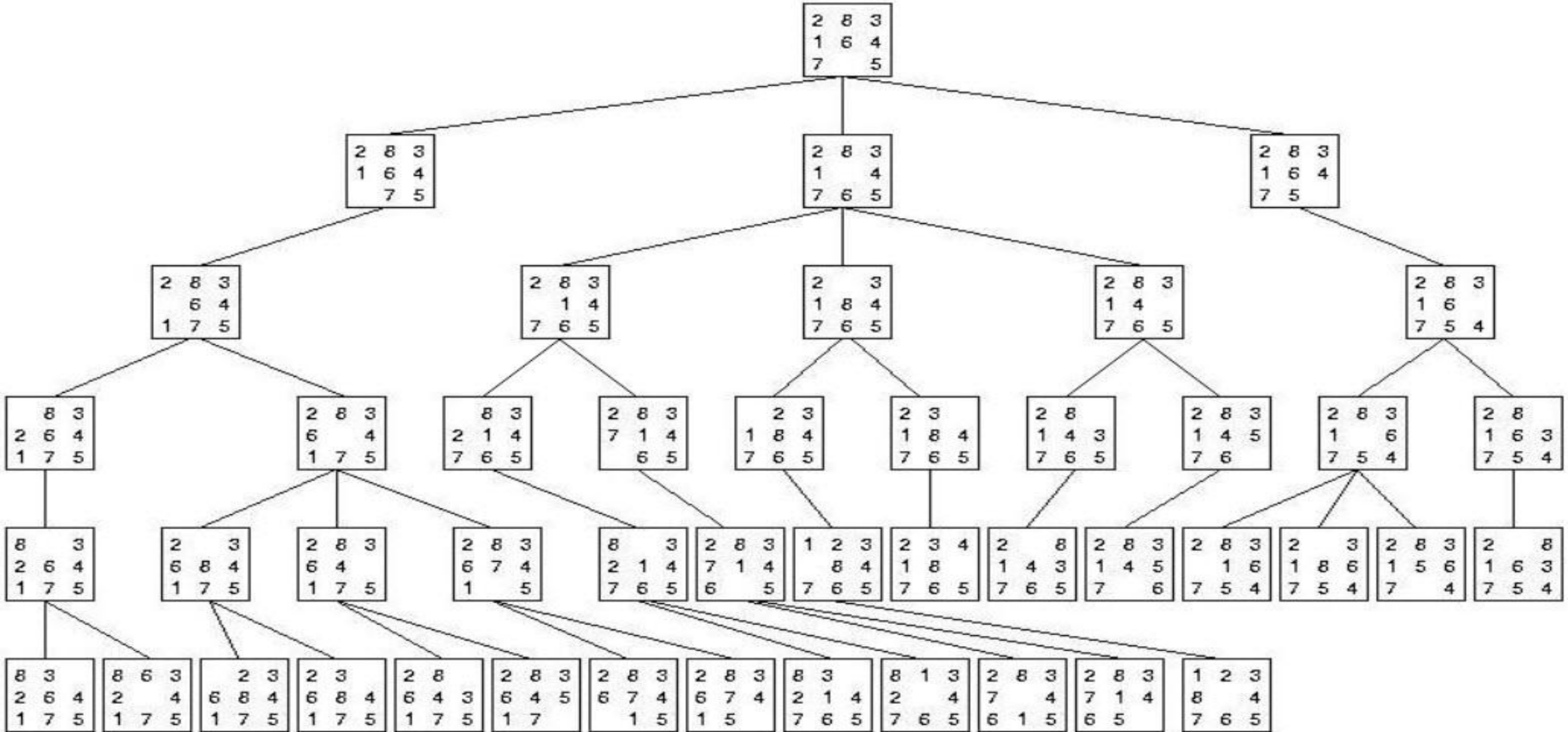
Sheet2 : Search A*

Eng. Abdulrahman ElMadkhoum

1. The following figure gives an example showing the BFS tree for the 8 puzzle problem. List the order of the visited nodes by the following search algorithms:
   a) Breadth first
   b) Depth First
   c) Limited Depth First with limit d=3.
   d) Best first Search using the misplaced tiles as a heuristic function.
   e) Best first Search using the Manhattan Distance as a heuristic function.
   f) A* Search using the misplaced tiles as a heuristic function.
   g) A* Search using the Manhattan Distance as a heuristic function.



2	8	3
1	6	4
7		5

Initial State

1	2	3
8		4
7	6	5

Goal State

f) A* using misplaced tiles as heuristic function

A* using misplaced tiles as heuristic function

Total Cost of a Node in A* = Heuristic + Cost

$f(n) = h(n) + g(n)$

First, Calculate Heuristic using Misplaced Tiles

2	8	3
1	6	4
7		5

Initial State

1 is misplaced
2 is misplaced
3 is ok
4 is ok
5 is ok
6 is misplaced
7 is ok
8 is misplaced
So, heuristic here = 1+1+1+1 =**4**

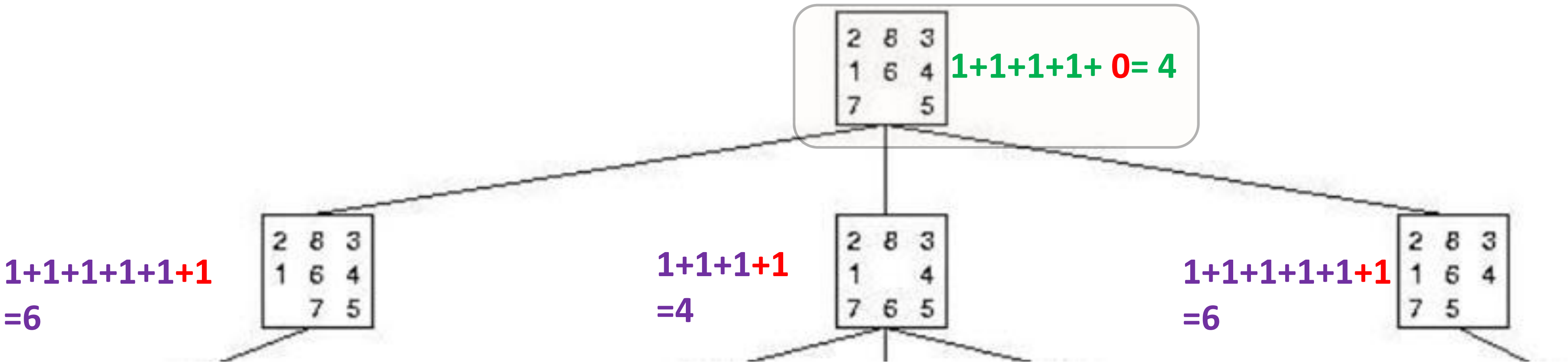
H(n)=4

1	2	3
8		4
7	6	5

Goal State

H(n) = 0

Then , Add the number of steps to the heuristic



The difference between Best first and A*

Best First

Total Cost of a Node

= Heuristic Only

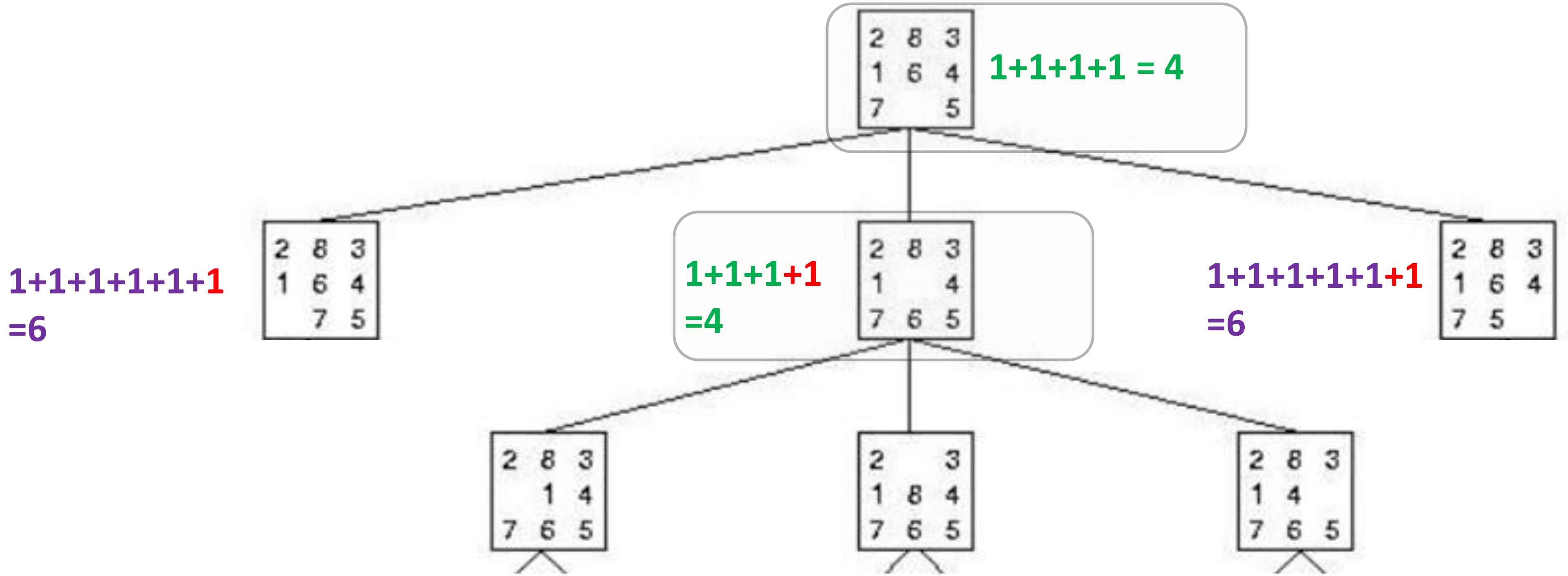
A*

Total Cost of a Node

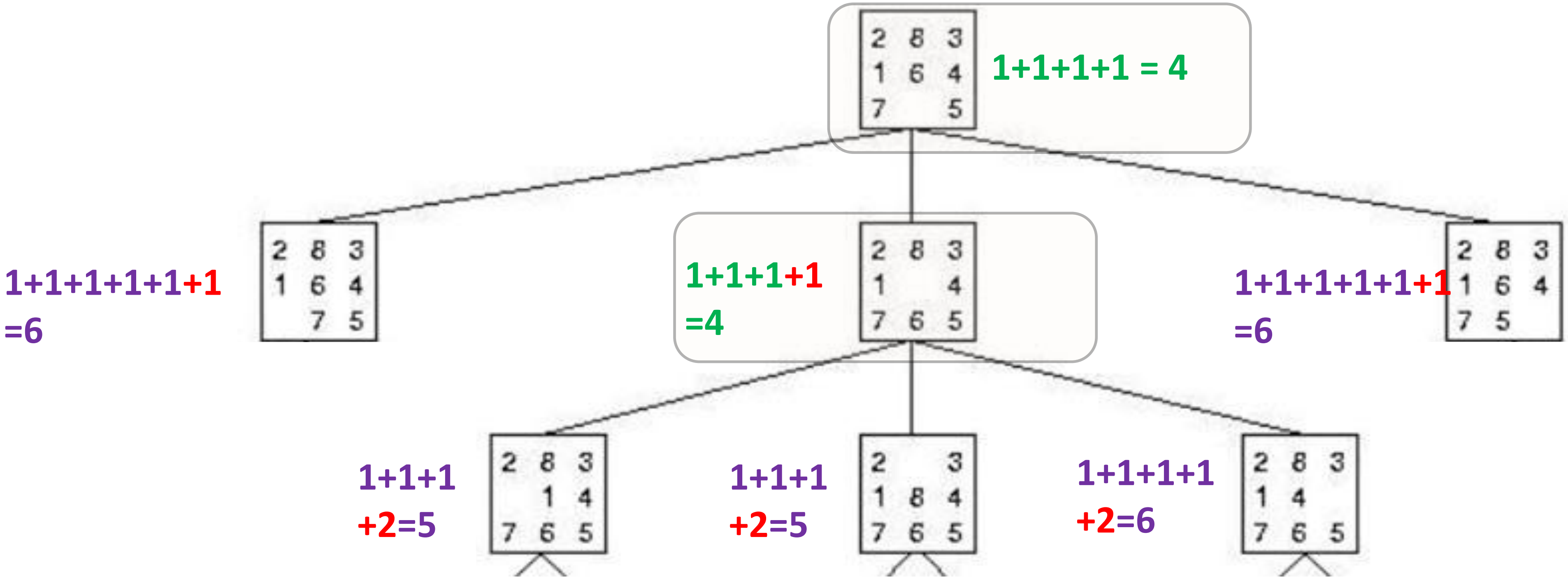
= Heuristic + Cost

Cost depends on the problem type
In 8 puzzle problem the cost to move from one node to another is 1

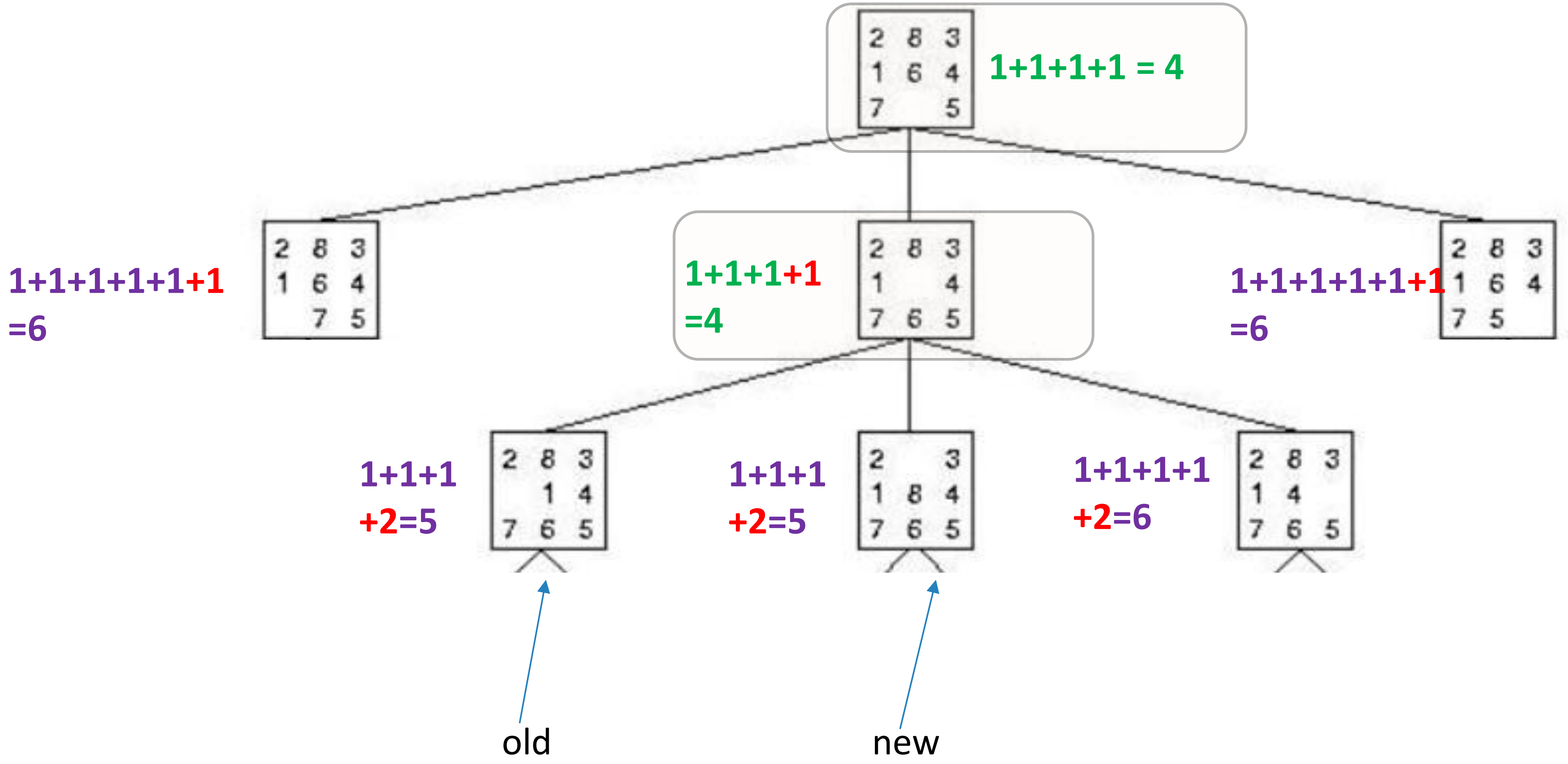
*Heuristic is calculated using **Misplaced** or **Manhattan** or other methods*



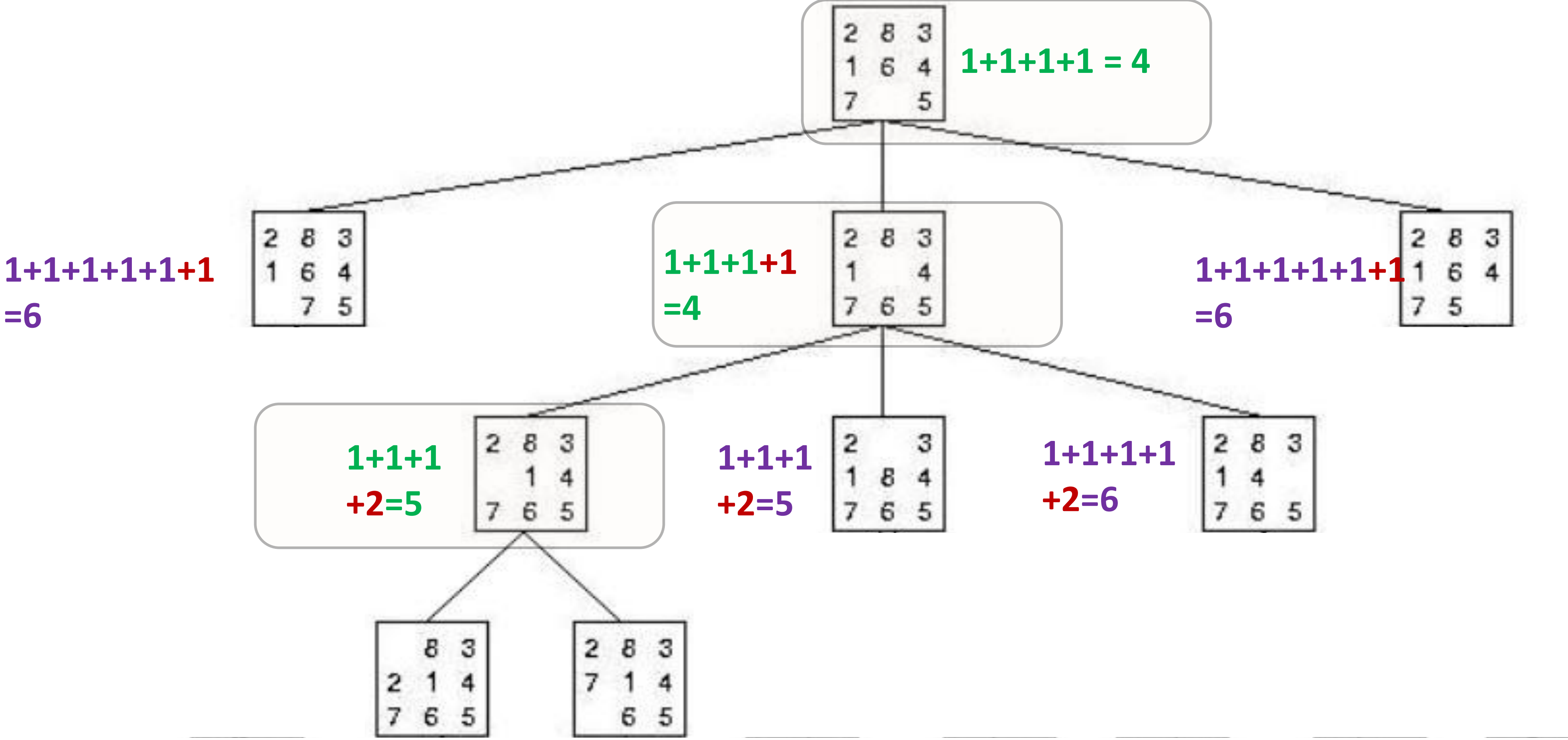
We Choose the Node with the least Heuristic + Cost in the whole tree

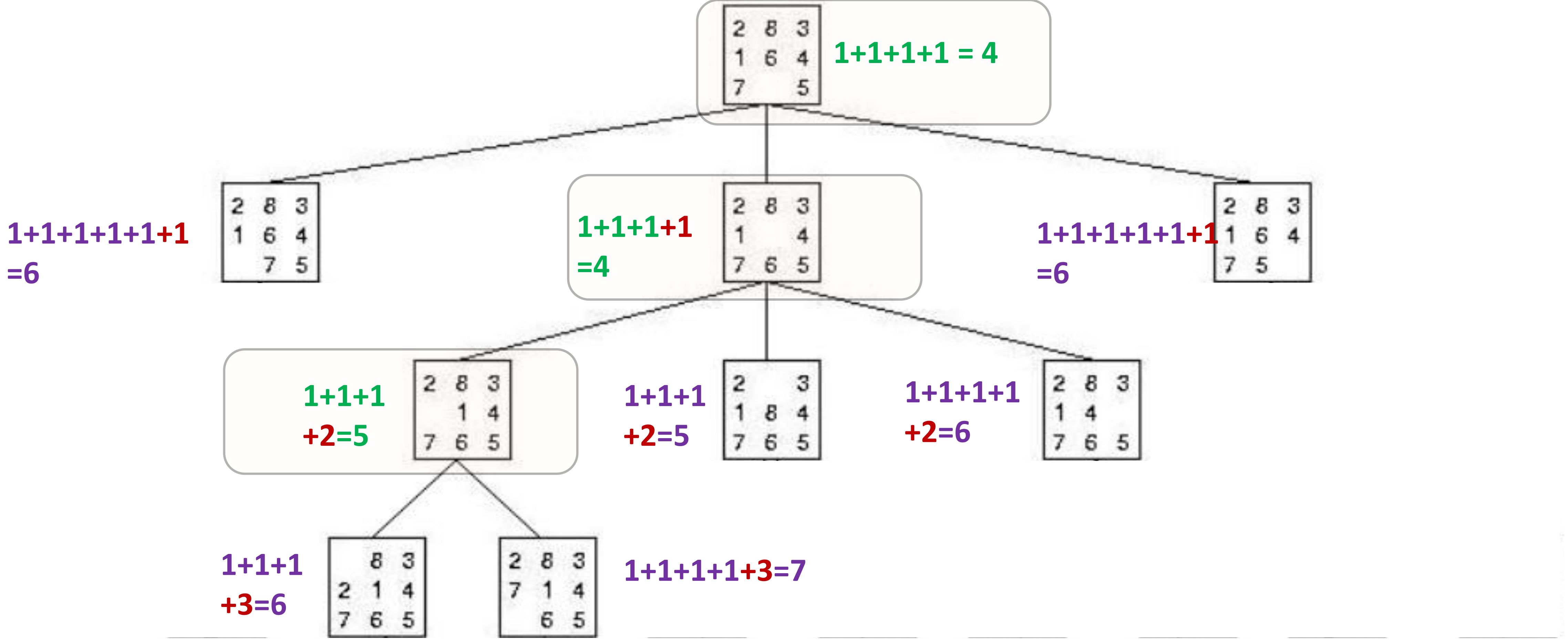


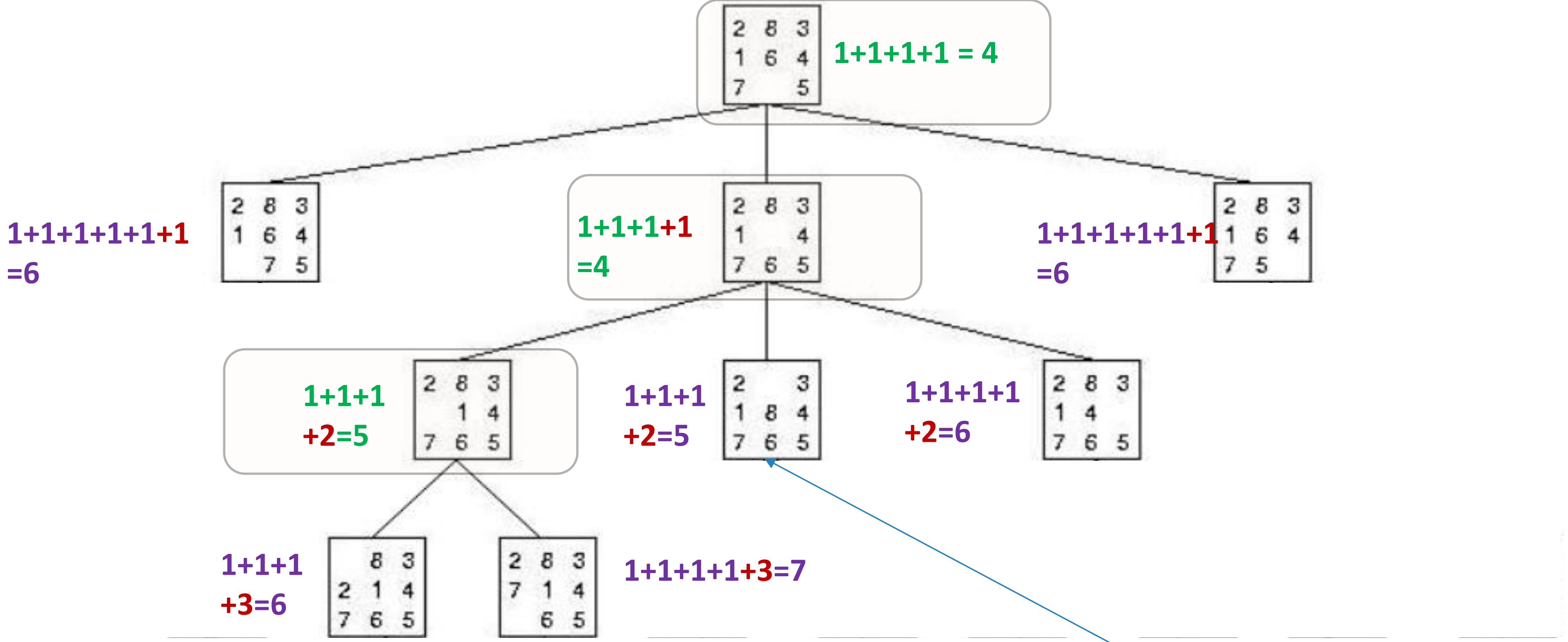
There are Two nodes with the same Total cost



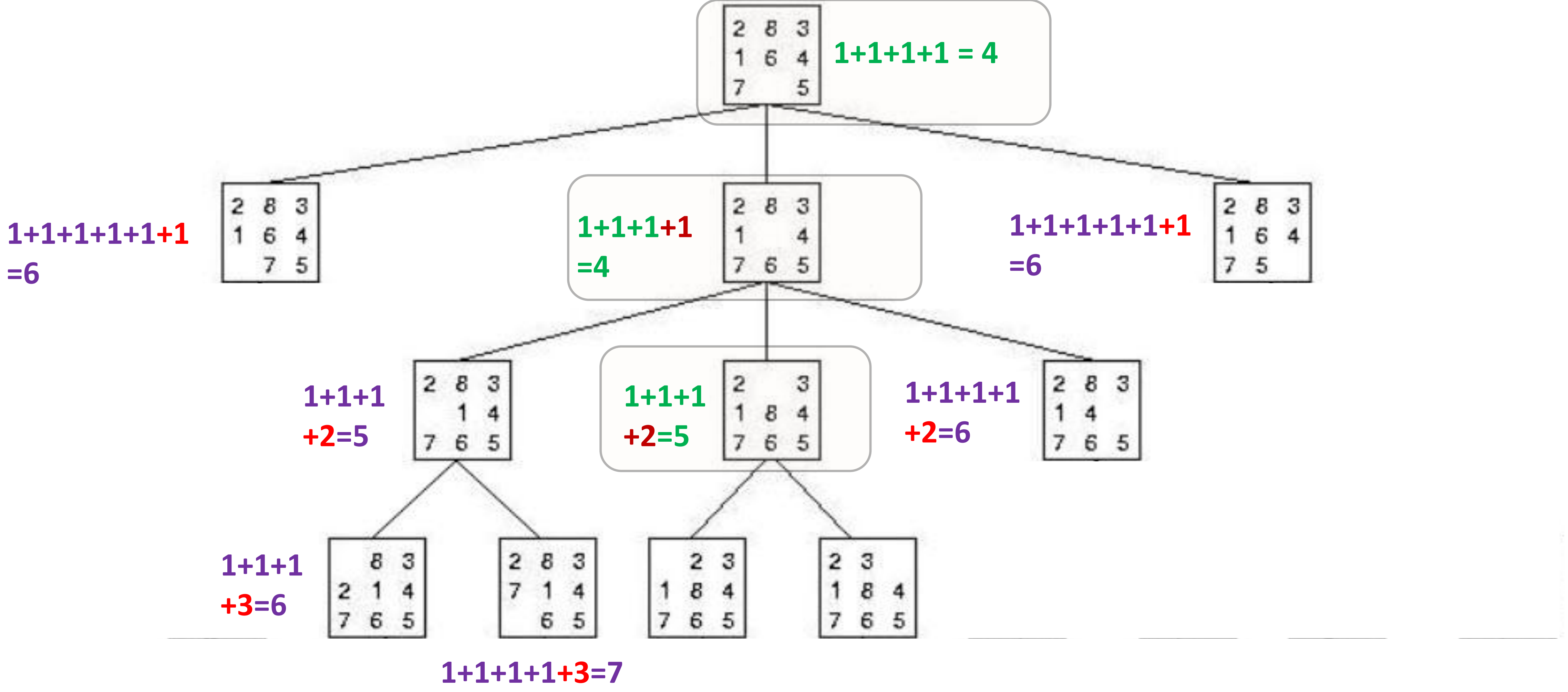
We Choose the Older one (The one we drew First)

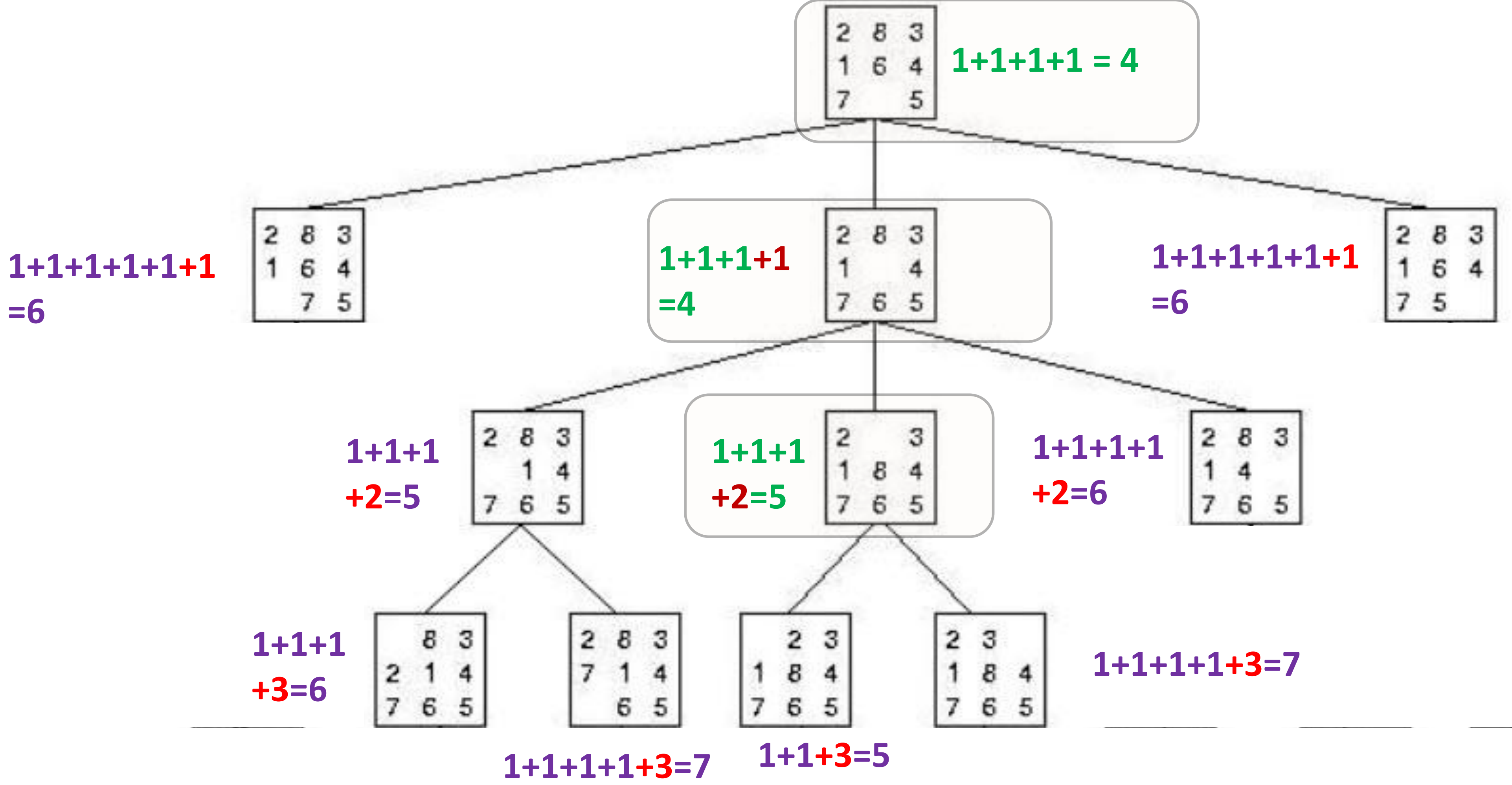

1+1+1+1 = 4
1+1+1+1+1+1
=6
1+1+1+1
=4
1+1+1+1+1+1
=6
1+1+1
+2=5
1+1+1
+2=5
1+1+1+1
+2=6


1+1+1+1 = 4
1+1+1+1+1+1
=6
1+1+1+1
=4
1+1+1+1+1+1
=6
1+1+1
+2=5
1+1+1
+2=5
1+1+1+1
+2=6
1+1+1
+3=6
1+1+1+1+3=7

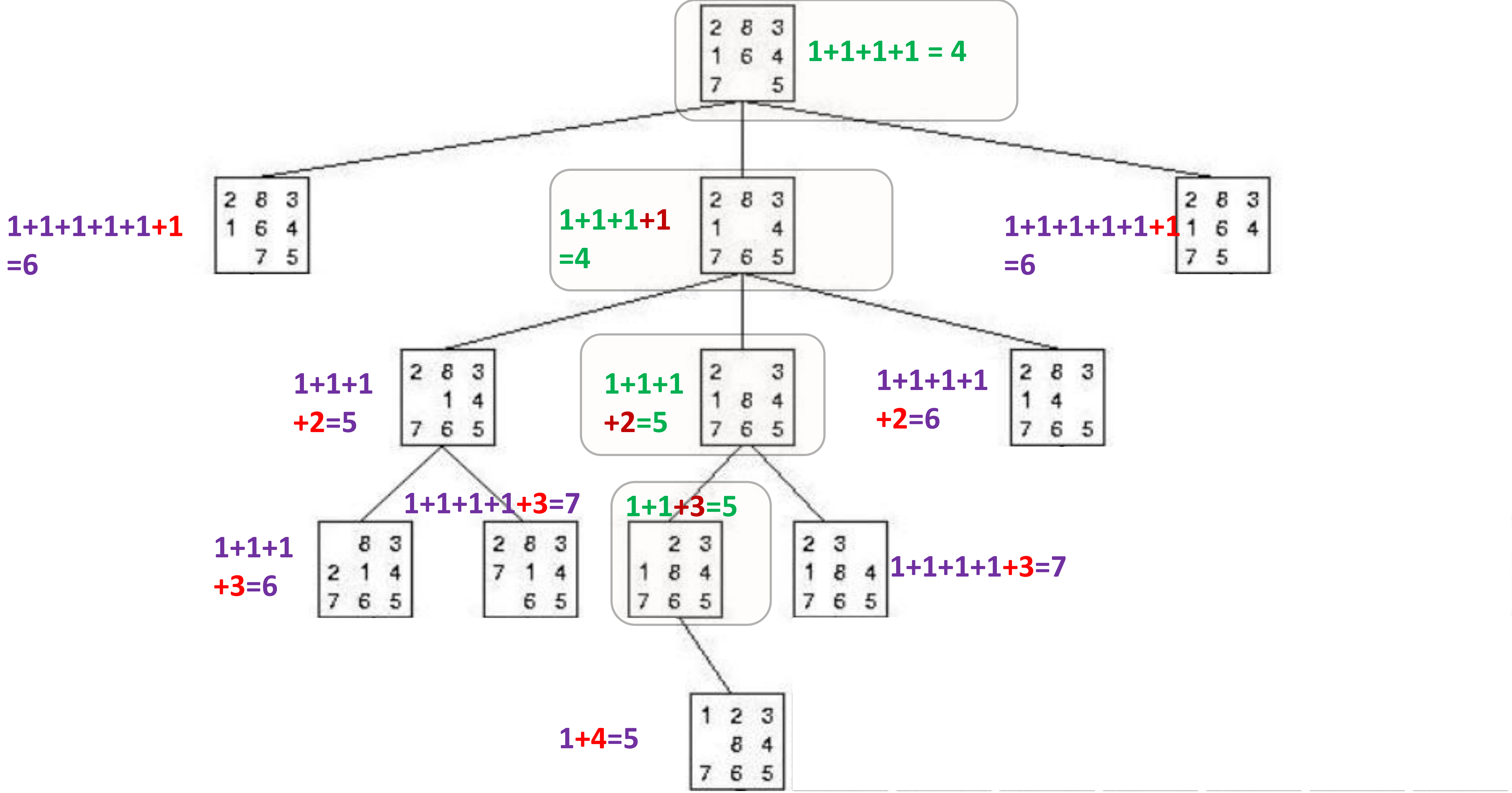


The Least Cost is not always at the current level , look at the previous levels

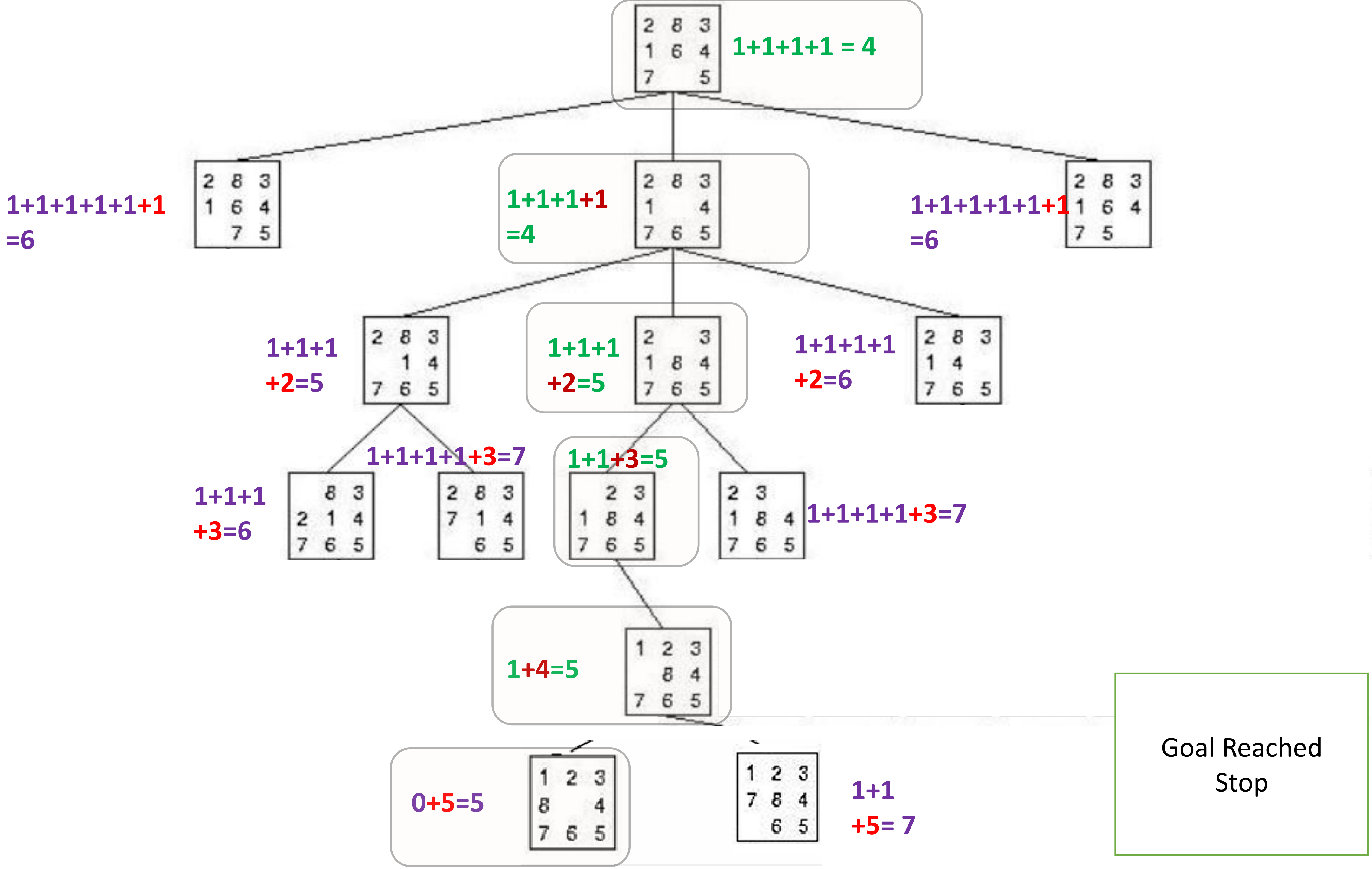

1+1+1+1 = 4
1+1+1+1+1+1
=6
1+1+1+1
=4
1+1+1+1+1+1
=6
1+1+1
+2=5
1+1+1
+2=5
1+1+1+1
+2=6
1+1+1
+3=6
1+1+1+1+3=7



1+1+1+1 = 4
1+1+1+1+1+1
=6
1+1+1+1
=4
1+1+1+1+1+1
=6
1+1+1
+2=5
1+1+1
+2=5
1+1+1+1
+2=6
1+1+1
+3=6
1+1+1+1+3=7
1+1+1+1+3=7
1+1+3=5



2 8 3
1 6 4
7 5
1+1+1+1 = 4
1+1+1+1+1+1
=6
2 8 3
1 6 4
7 5
1+1+1+1
=4
2 8 3
1 4
7 6 5
1+1+1+1+1+1
=6
2 8 3
1 6 4
7 5
1+1+1
+2=5
2 8 3
1 4
7 6 5
1+1+1
+2=5
2 3
1 8 4
7 6 5
1+1+1+1
+2=6
2 8 3
1 4
7 6 5
1+1+1
+3=6
8 3
2 1 4
7 6 5
1+1+1+1+3=7
2 8 3
7 1 4
6 5
1+1+3=5
2 3
1 8 4
7 6 5
2 3
1 8 4
7 6 5
1+1+1+1+3=7
1+4=5
1 2 3
8 4
7 6 5


1+1+1+1 = 4
1+1+1+1+1+1
=6
1+1+1+1
=4
1+1+1+1+1+1
=6
1+1+1
+2=5
1+1+1
+2=5
1+1+1+1
+2=6
1+1+1+1+3=7
1+1+3=5
1+1+1
+3=6
1+1+1+1+3=7
1+4=5
0+5=5
1+1
+5= 7
Goal Reached
Stop

g) A* using Manhattan Distance as heuristic function

First, Calculate Heuristic using Manhattan Distance

2	8	3
1	6	4
7		5

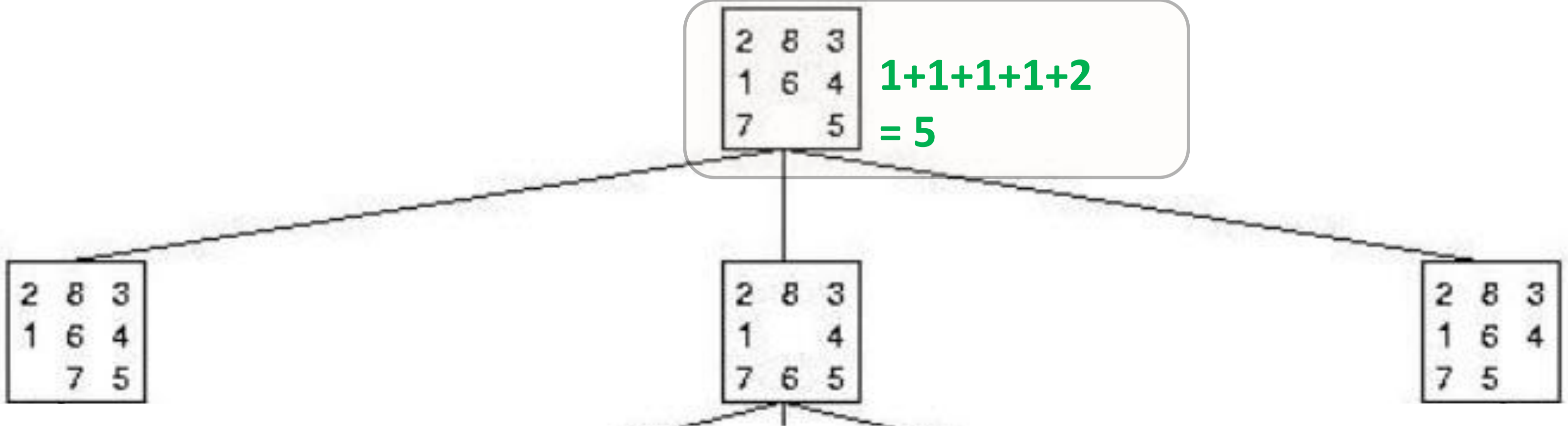
Initial State

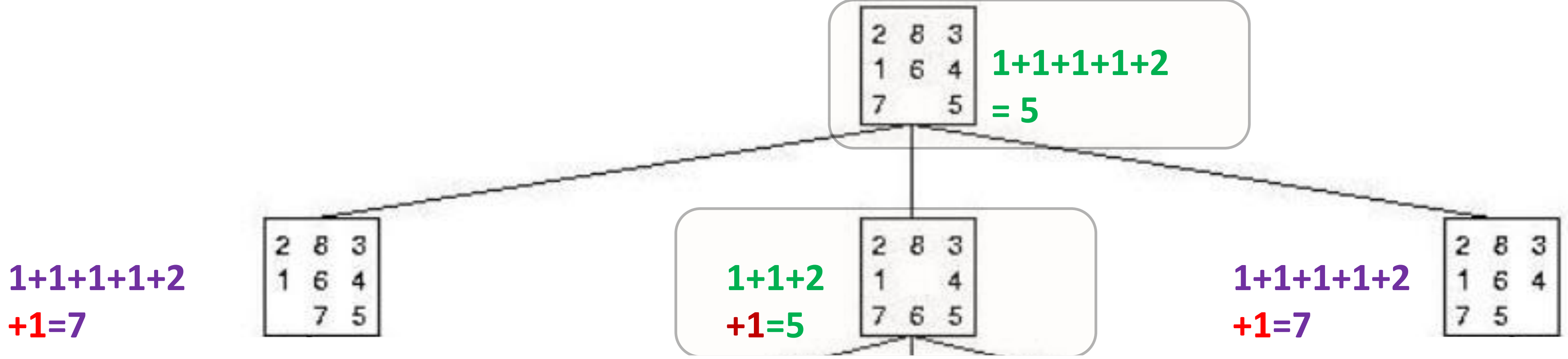
1	2	3
8		4
7	6	5

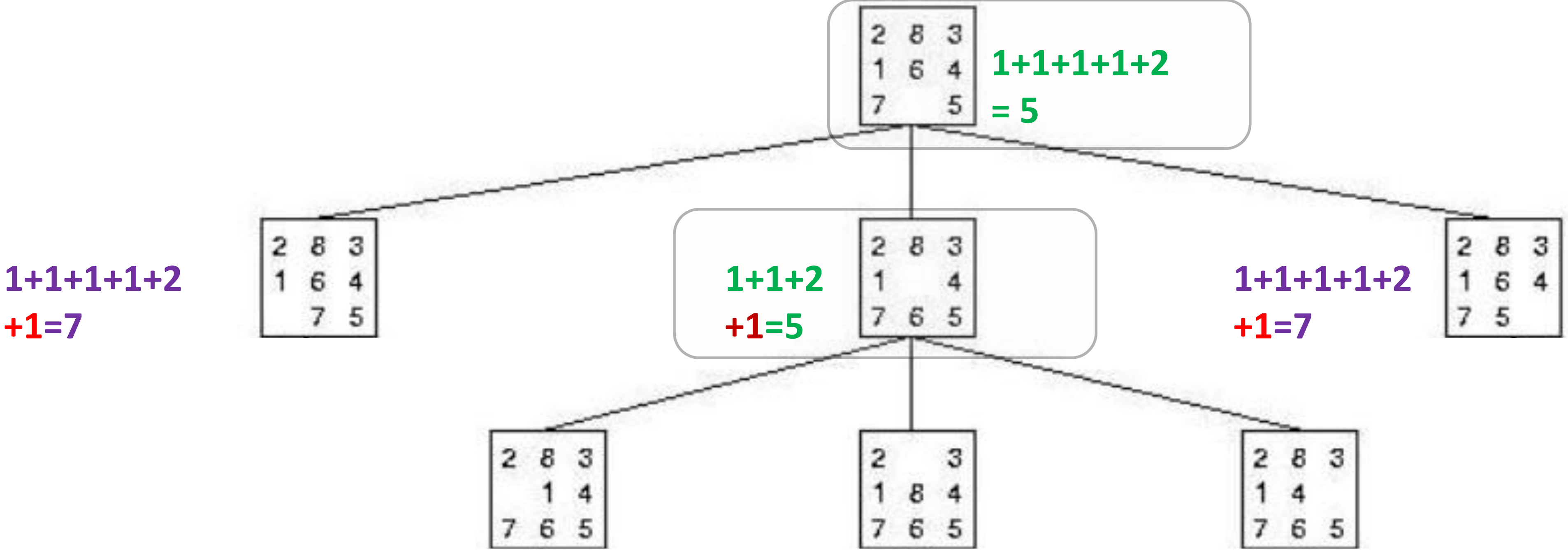
1 needs 1 movement to be in it’s correct position
2 needs 1 movement to be in it’s correct position
3 is in it’s correct position
4 is in it’s correct position
5 is in it’s correct position
6 needs 1 movement to be in it’s correct position
7 is in it’s correct position
8 needs 2 movements to be in it’s correct position
So, The Heuristics is = 1+1+1+2 = 5

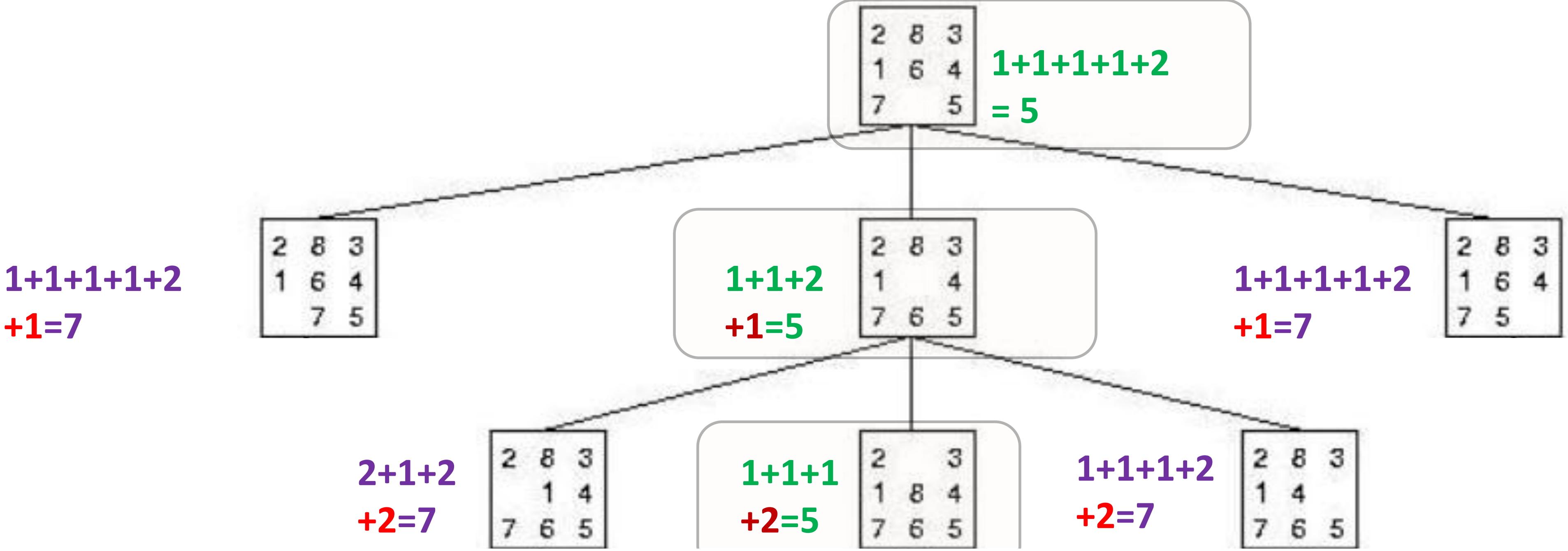
H(n) = 0

Then , Add the number of steps
to the heuristic

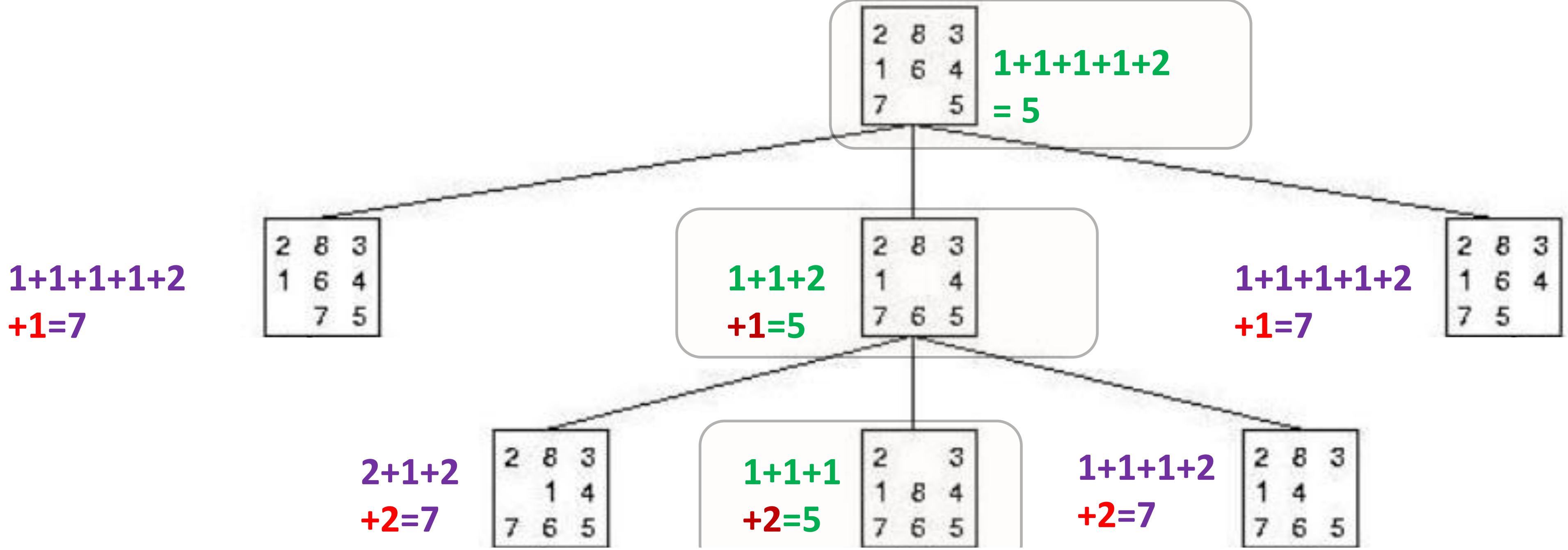

1+1+1+1+2
= 5


2 8 3
1 6 4
7 5
1+1+1+1+2
= 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
1+1+2
+1=5
2 8 3
1 4
7 6 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5

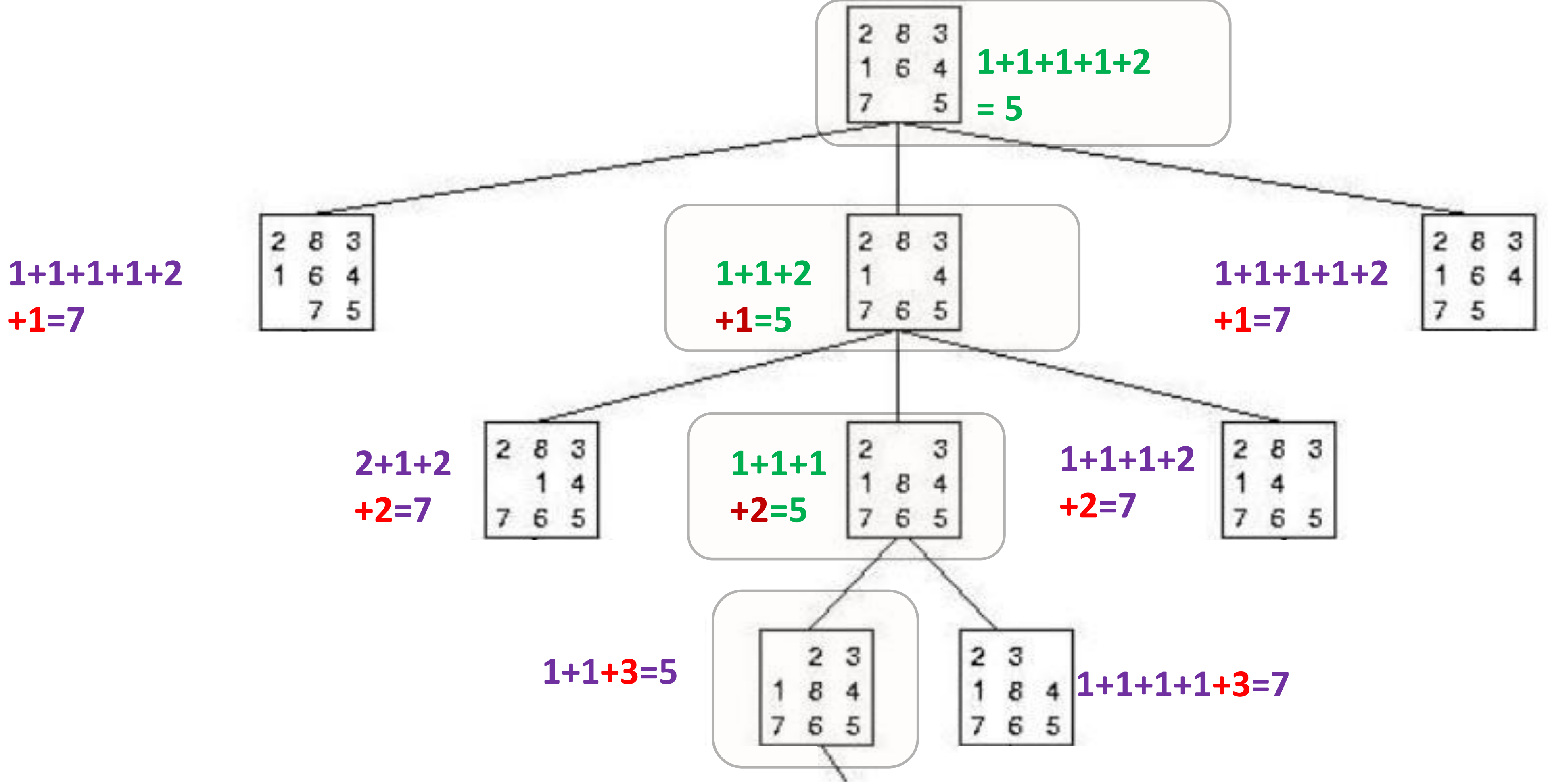

2 8 3
1 6 4
7 5
1+1+1+1+2
= 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
1+1+2
+1=5
2 8 3
1 4
7 6 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
2 8 3
1 4
7 6 5
2 3
1 8 4
7 6 5
2 8 3
1 4
7 6 5



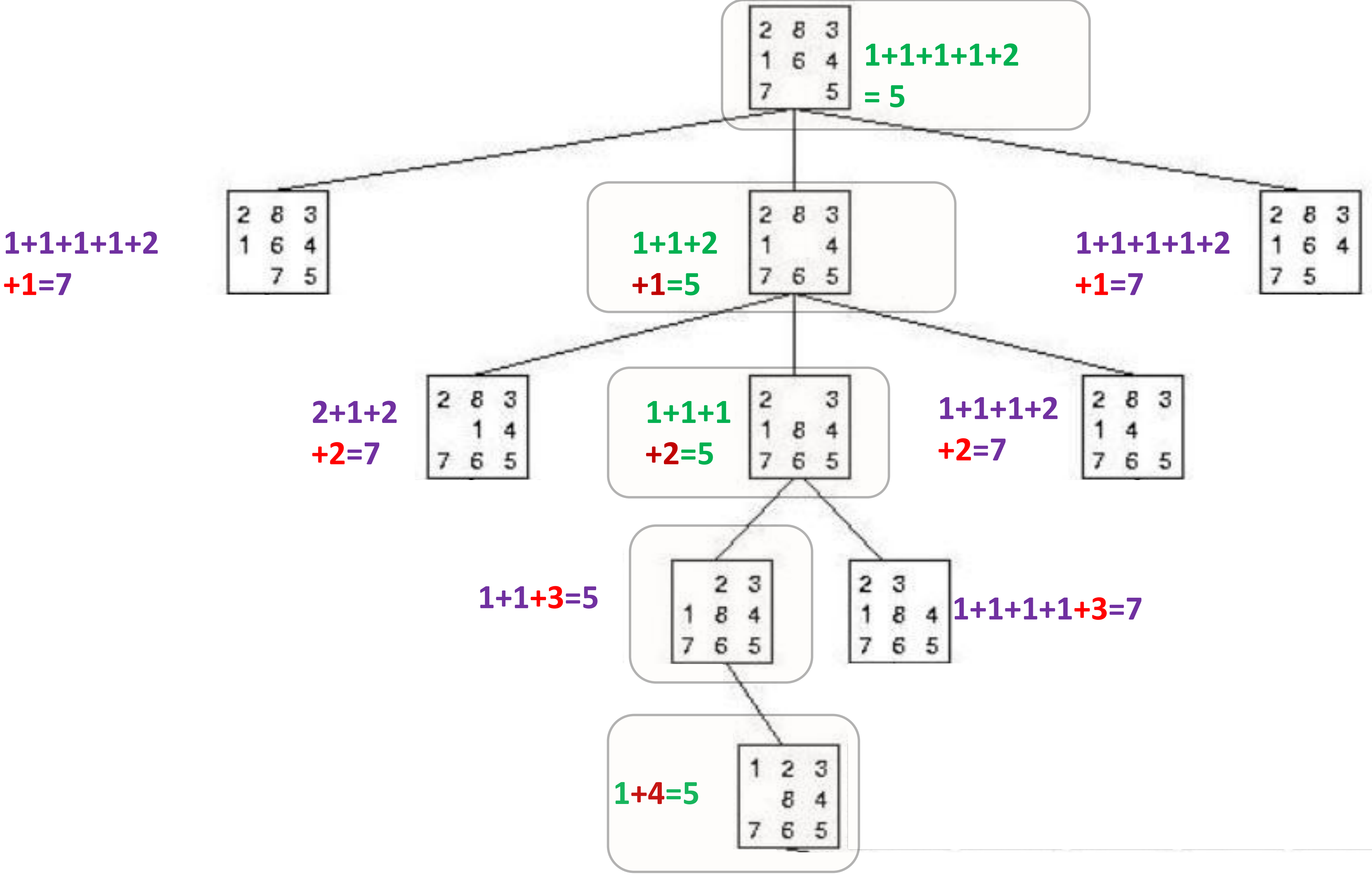
2 8 3
1 6 4
7 5
1+1+1+1+2
= 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
1+1+2
+1=5
2 8 3
1 4
7 6 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
2+1+2
+2=7
2 8 3
1 4
7 6 5
1+1+1
+2=5
2 3
1 8 4
7 6 5
1+1+1+2
+2=7
2 8 3
1 4
7 6 5



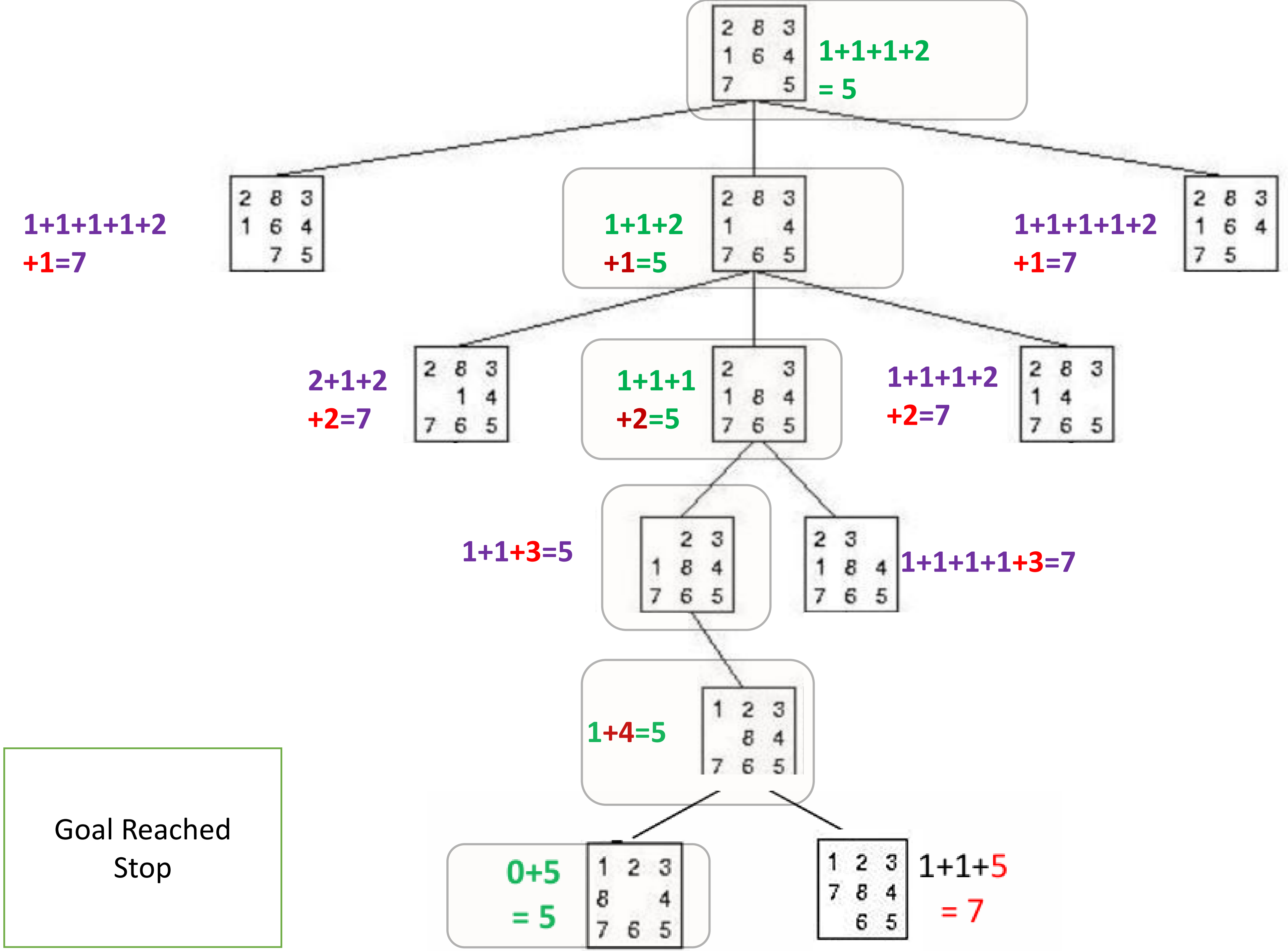
2 8 3
1 6 4
7 5
1+1+1+1+2
= 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
1+1+2
+1=5
2 8 3
1 4
7 6 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
2+1+2
+2=7
2 8 3
1 4
7 6 5
1+1+1
+2=5
2 3
1 8 4
7 6 5
1+1+1+2
+2=7
2 8 3
1 4
7 6 5



2 8 3
1 6 4
7 5
1+1+1+1+2
= 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
1+1+2
+1=5
2 8 3
1 4
7 6 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5
2+1+2
+2=7
2 8 3
1 4
7 6 5
1+1+1
+2=5
2 3
1 8 4
7 6 5
1+1+1+2
+2=7
2 8 3
1 4
7 6 5
1+1+3=5
2 3
1 8 4
7 6 5
2 3
1 8 4
7 6 5
1+1+1+1+3=7



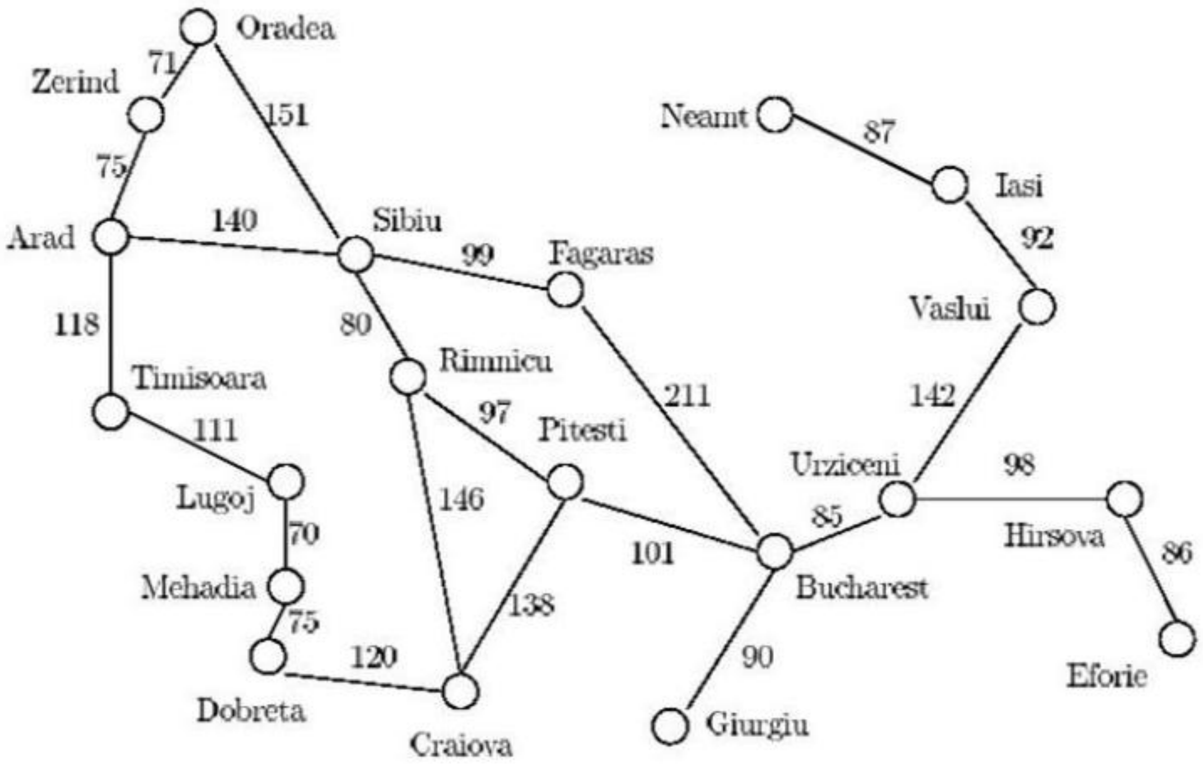
2 8 3
1 6 4
7 _ 5
1+1+1+1+2
= 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
_ 7 5
1+1+2
+1=5
2 8 3
1 _ 4
7 6 5
1+1+1+1+2
+1=7
2 8 3
1 6 4
7 5 _
2+1+2
+2=7
2 8 3
_ 1 4
7 6 5
1+1+1
+2=5
2 _ 3
1 8 4
7 6 5
1+1+1+2
+2=7
2 8 3
1 4 _
7 6 5
1+1+3=5
_ 2 3
1 8 4
7 6 5
2 3 _
1 8 4
7 6 5
1+1+1+1+3=7
1+4=5
1 2 3
_ 8 4
7 6 5


1+1+1+2
= 5
1+1+1+1+2
+1=7
1+1+2
+1=5
1+1+1+1+2
+1=7
2+1+2
+2=7
1+1+1
+2=5
1+1+1+2
+2=7
1+1+3=5
1+1+1+1+3=7
1+4=5
Goal Reached
Stop
0+5
= 5
1+1+5
= 7

Question 2

2. Given the Arad/Bucharest route finding problem and the heuristics table below, show the list of visited nodes using the following search techniques:
   a. Breadth first
   b. Depth First
   c. Best first.
   d. A* Search

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374



In this question the heuristic value is given in the table and the cost values are in the map

$f(n) = H(n) + g(n)$

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
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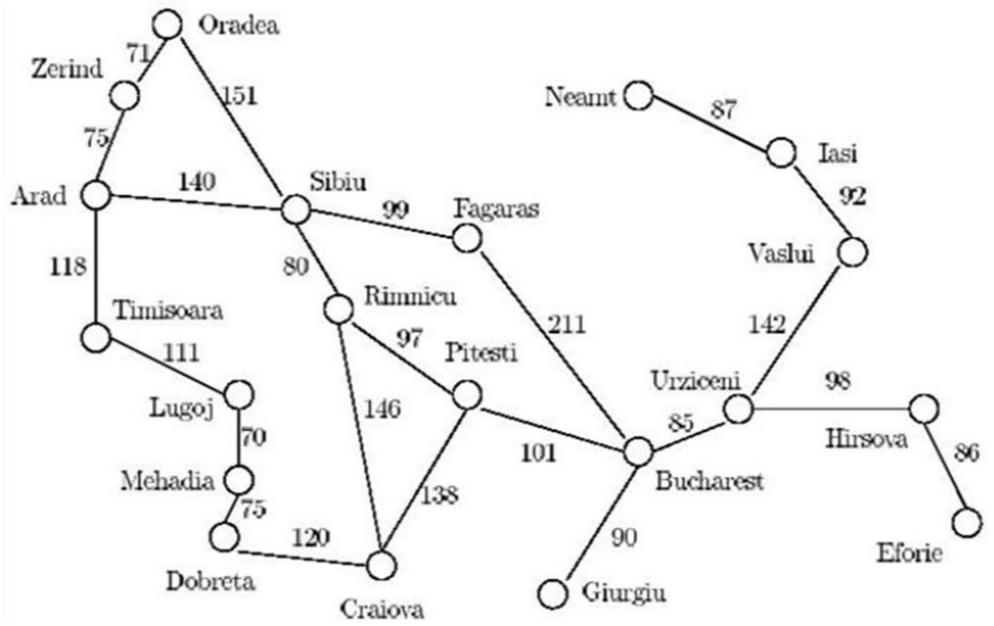

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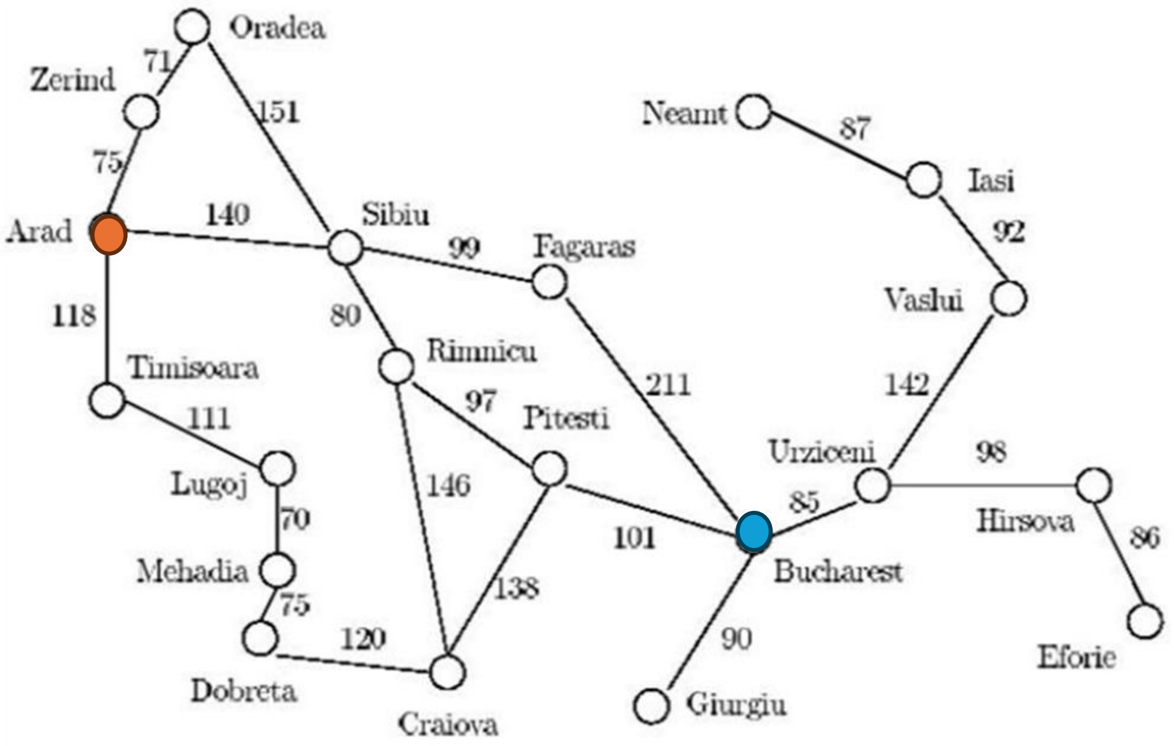
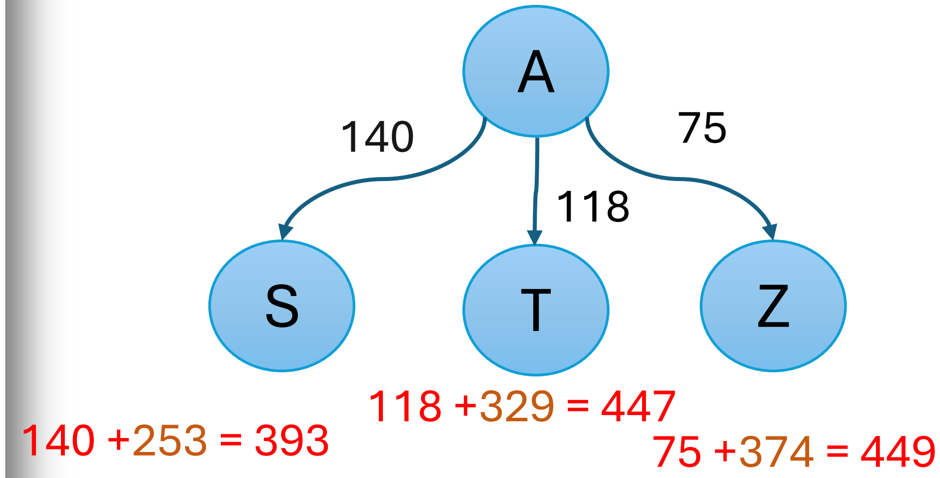


A*



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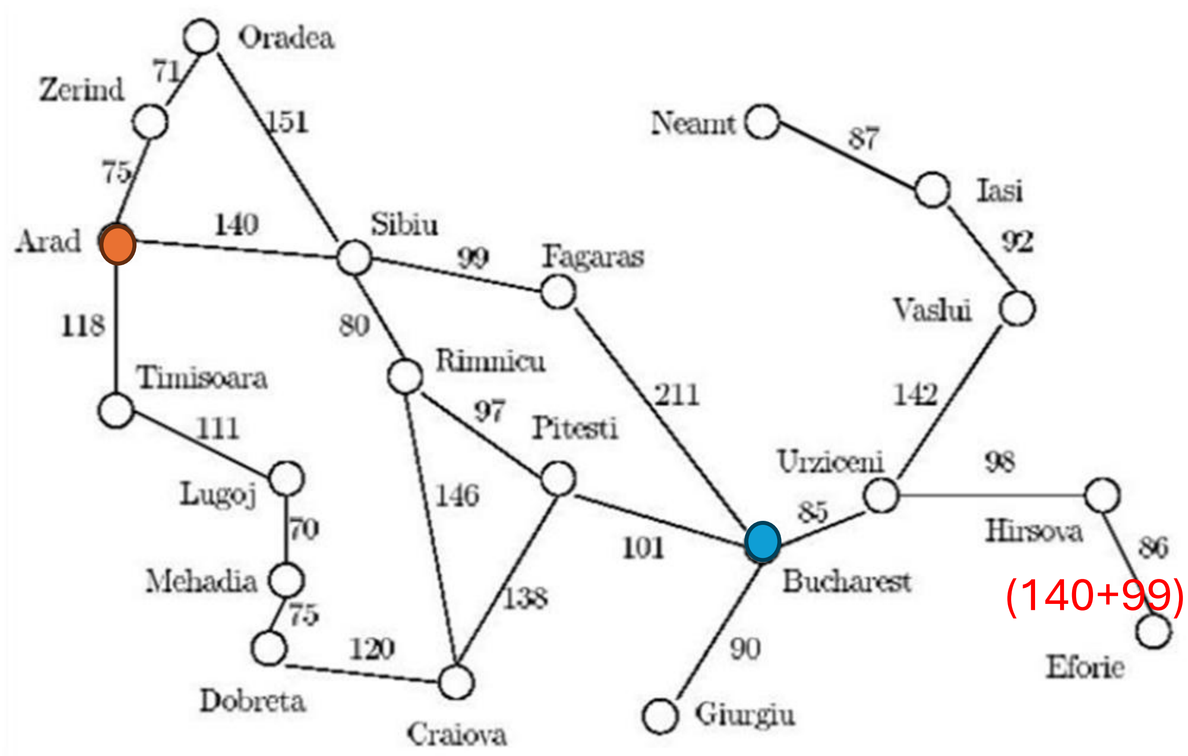
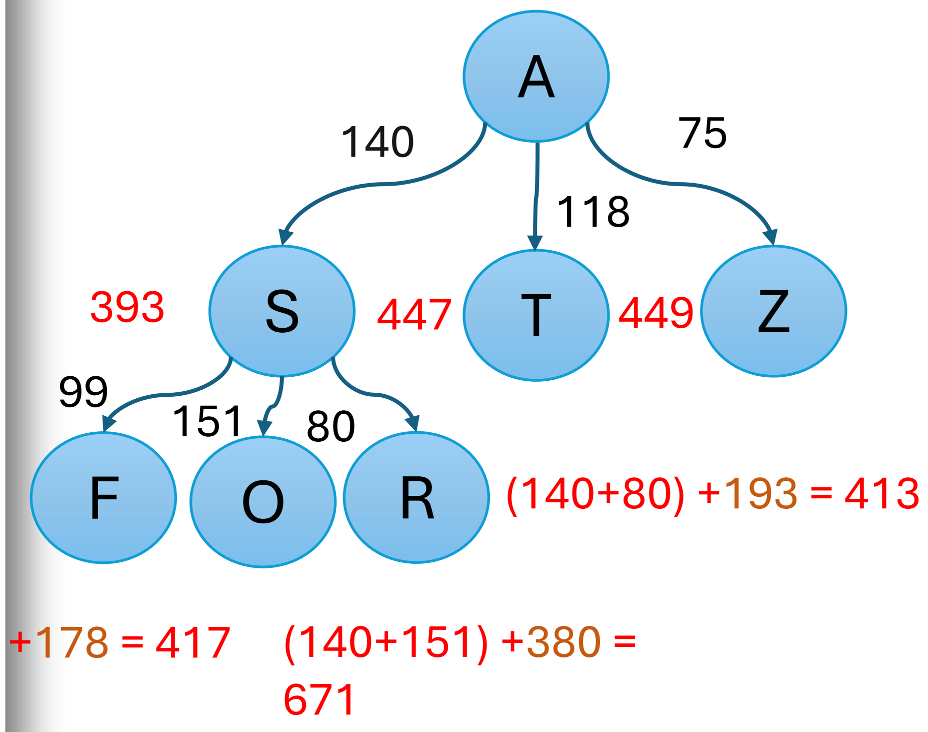


A*



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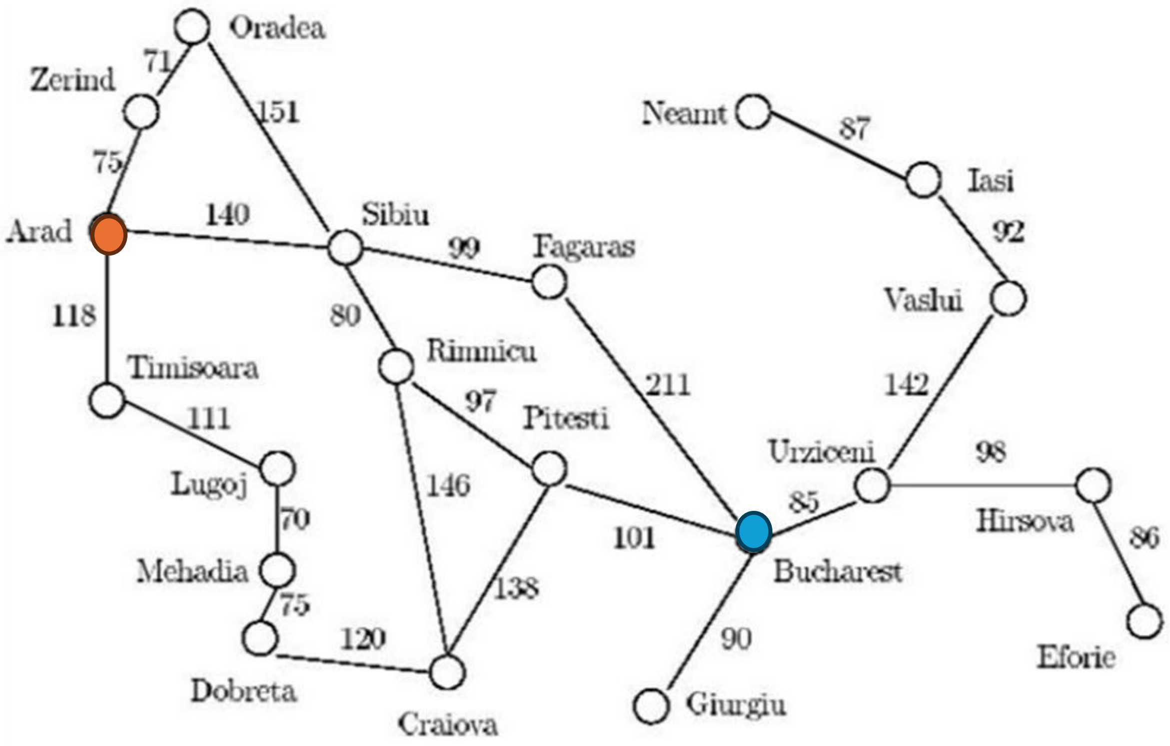

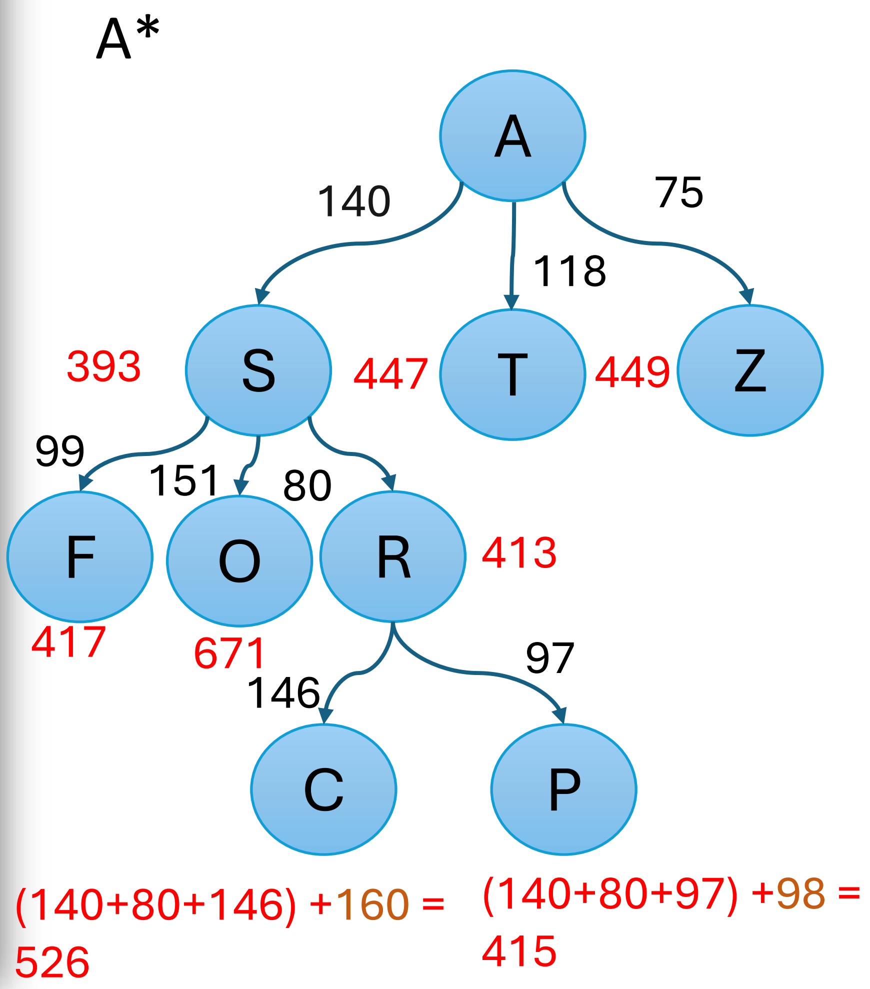

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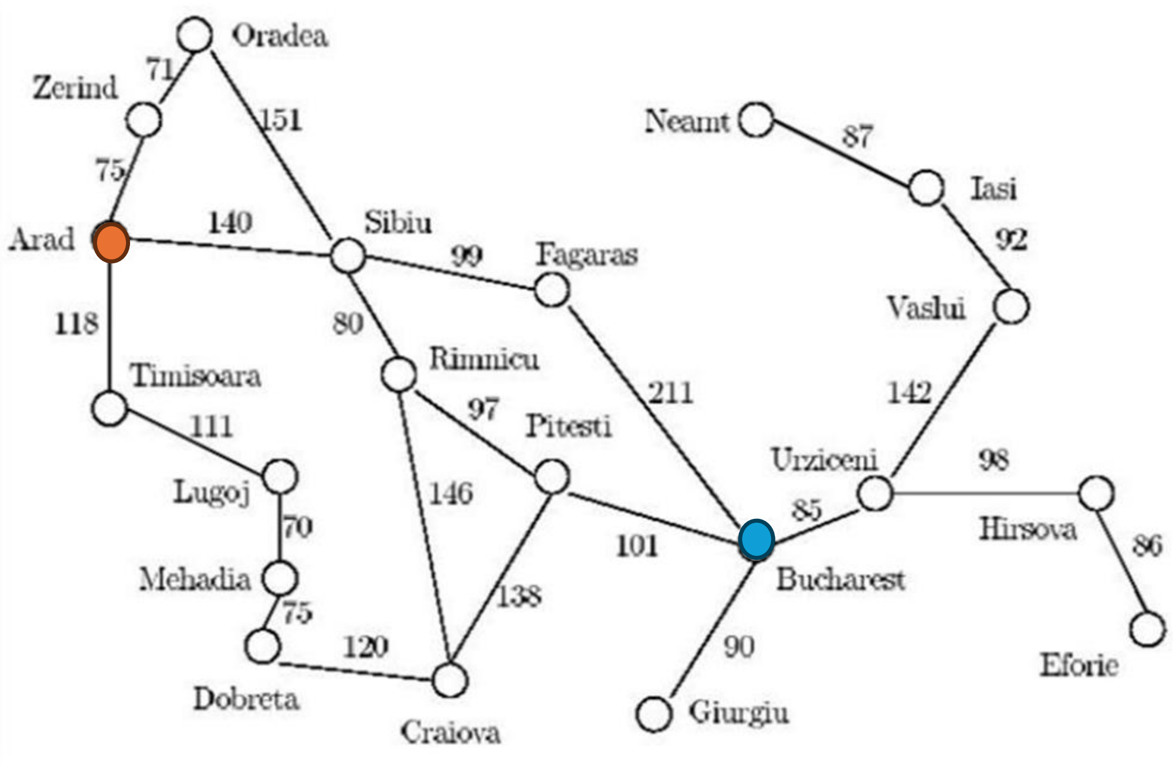

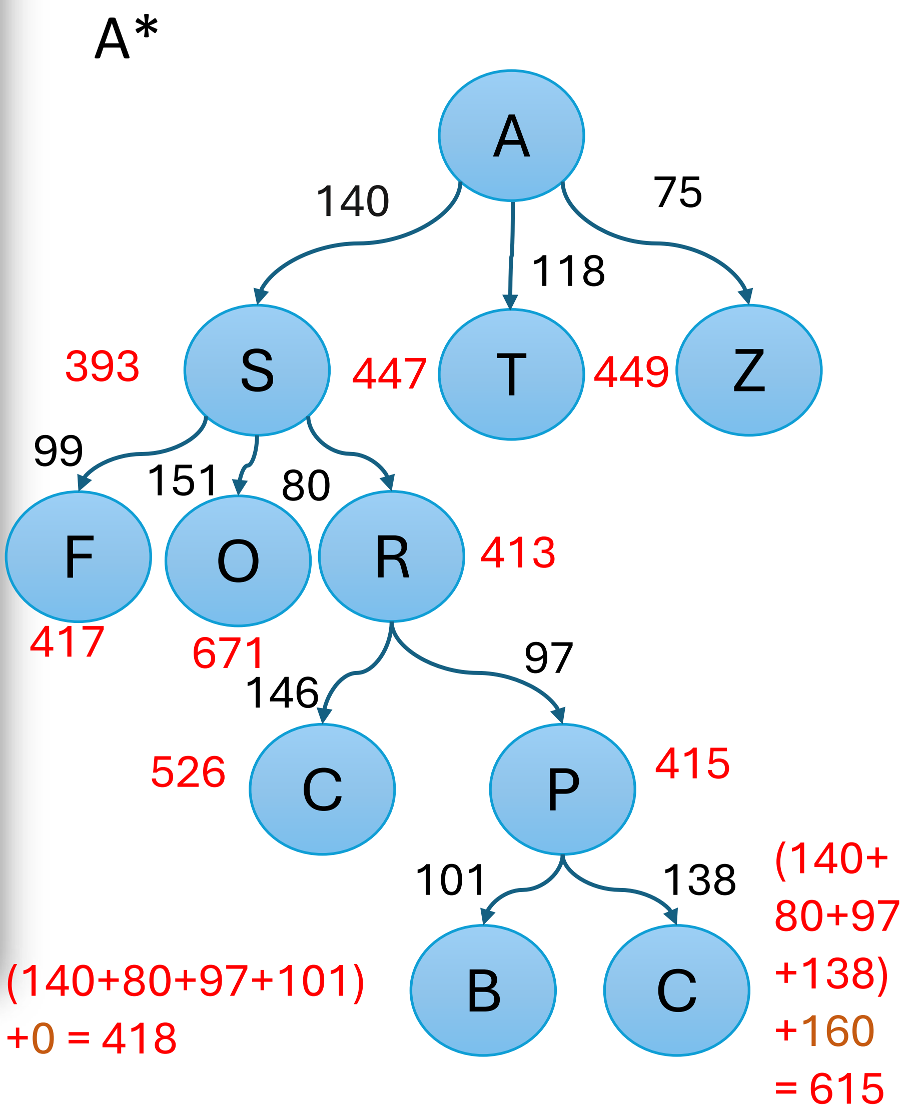

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Oradea	380
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Rimnicu Vilcea	193
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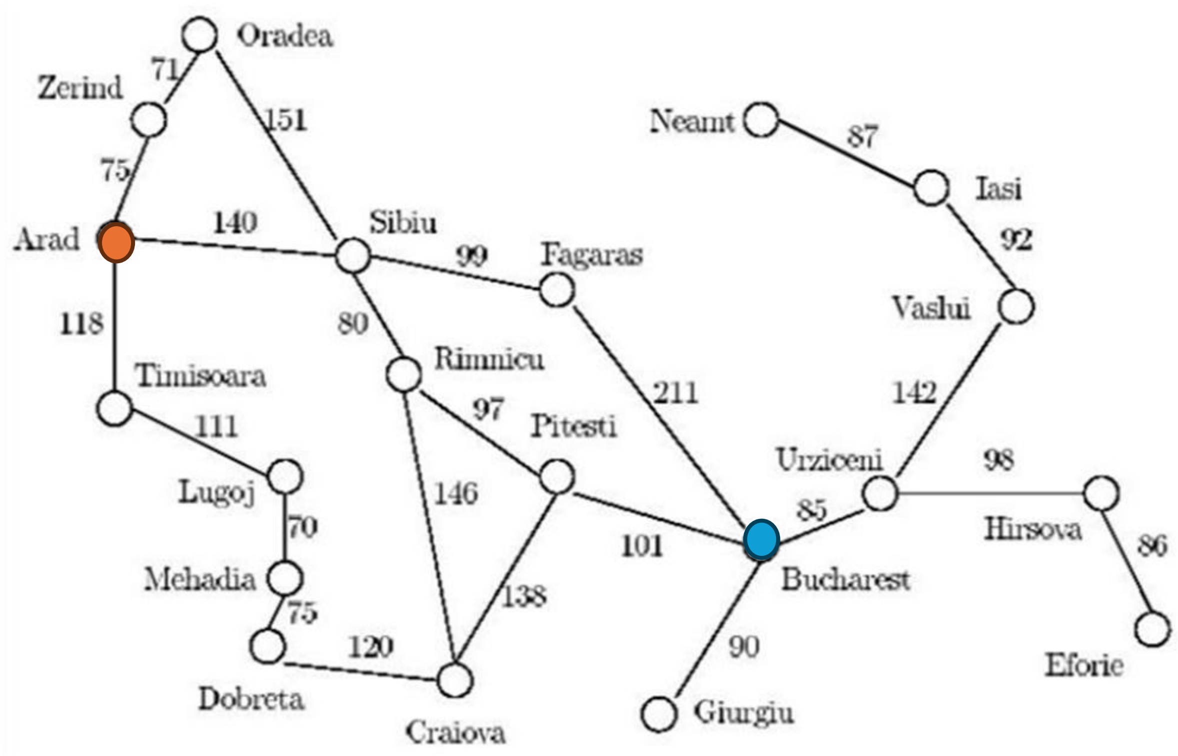
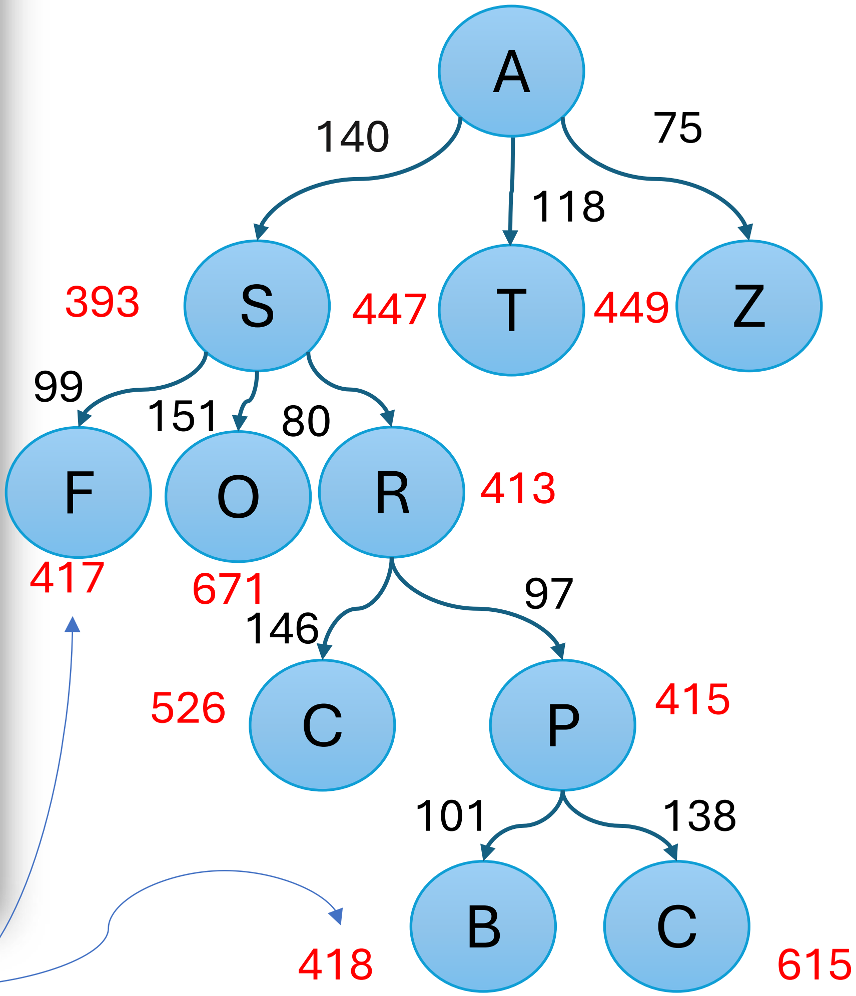


A*



Reached Goal but it doesn't have the least Total Cost

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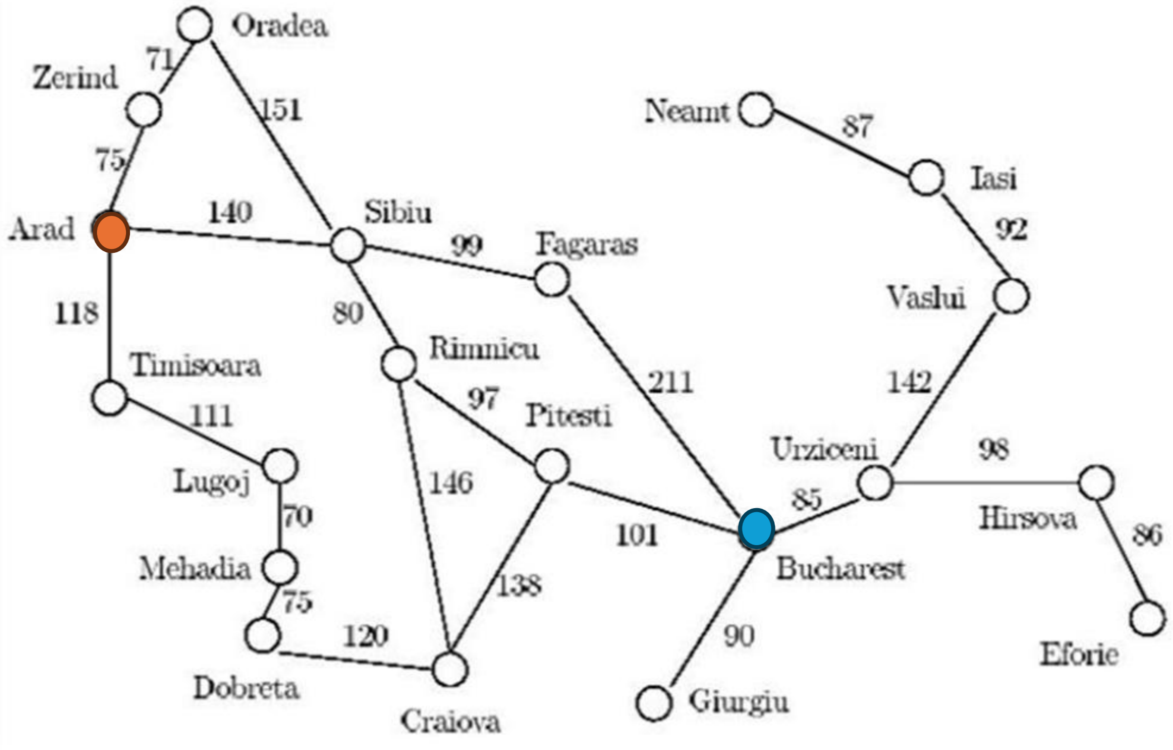
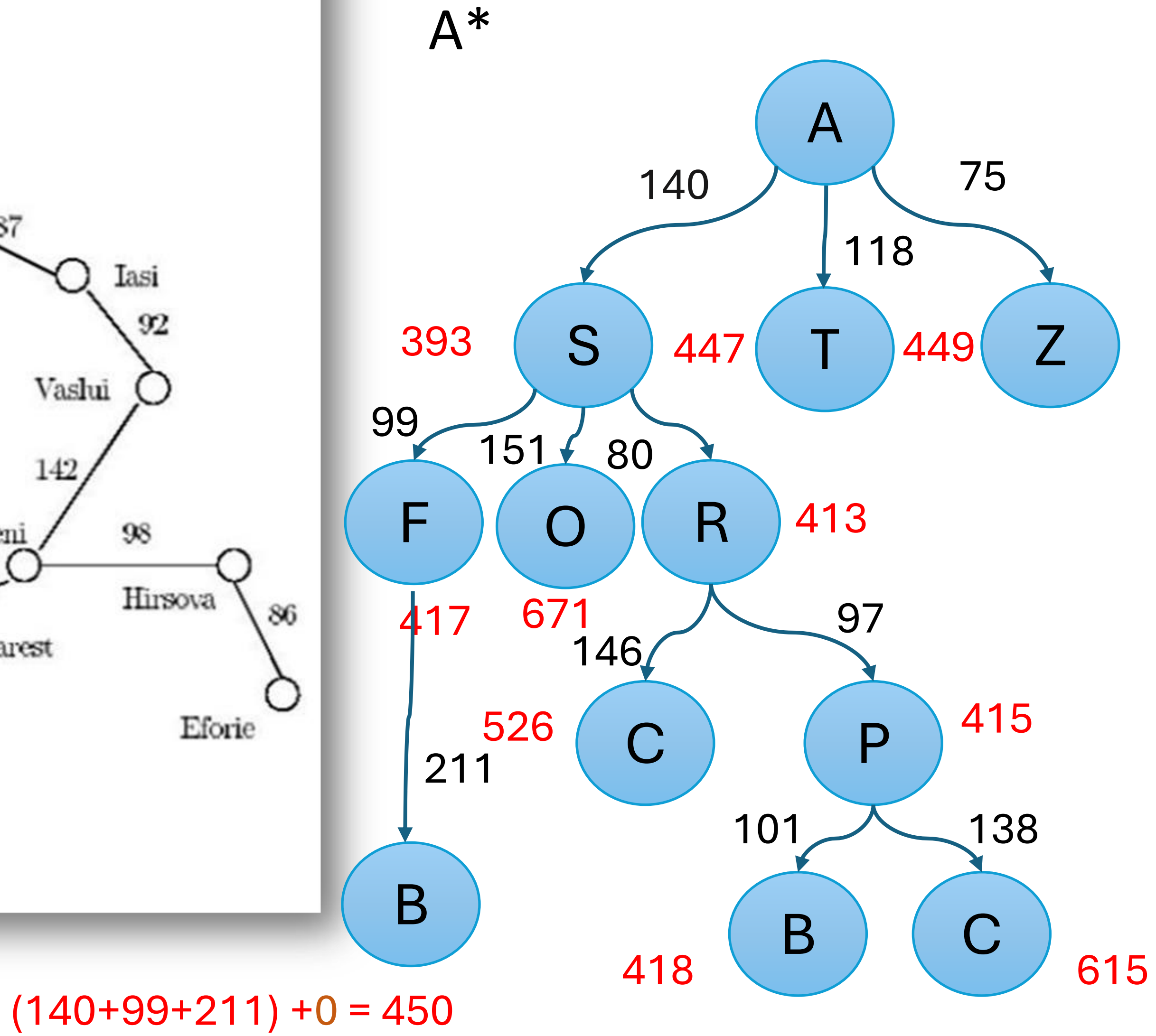
Zerind	374

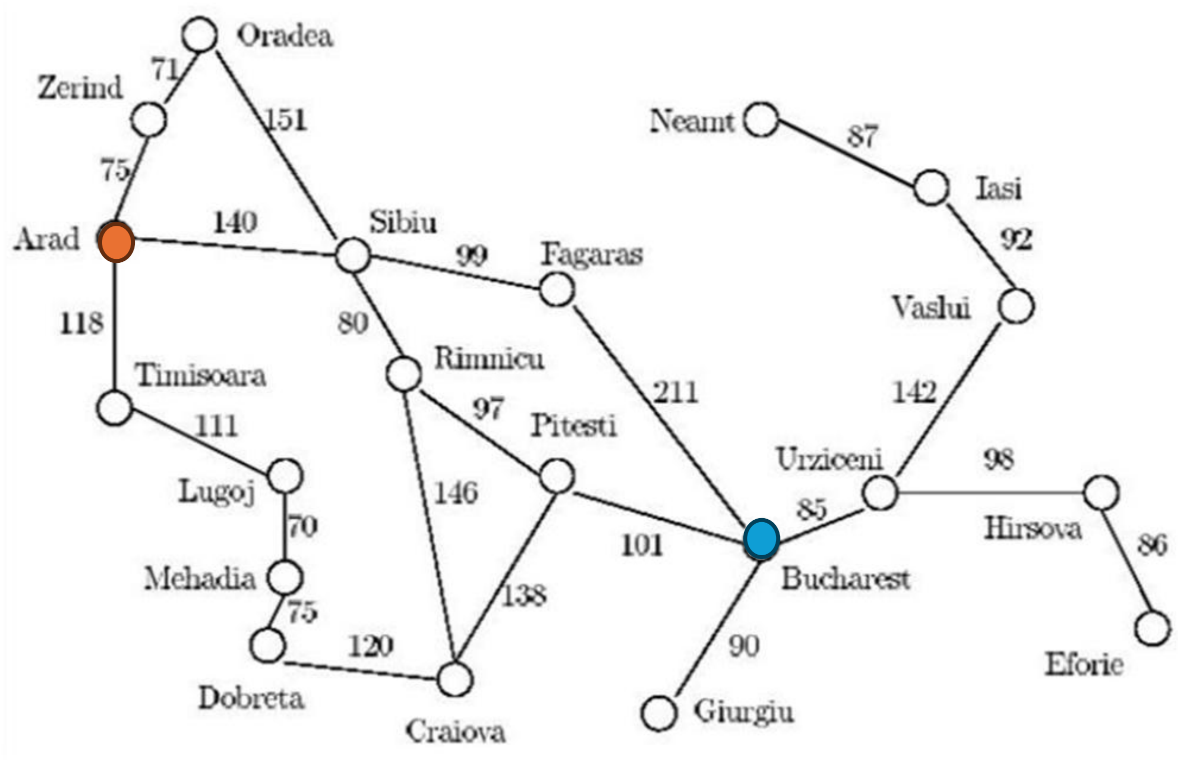




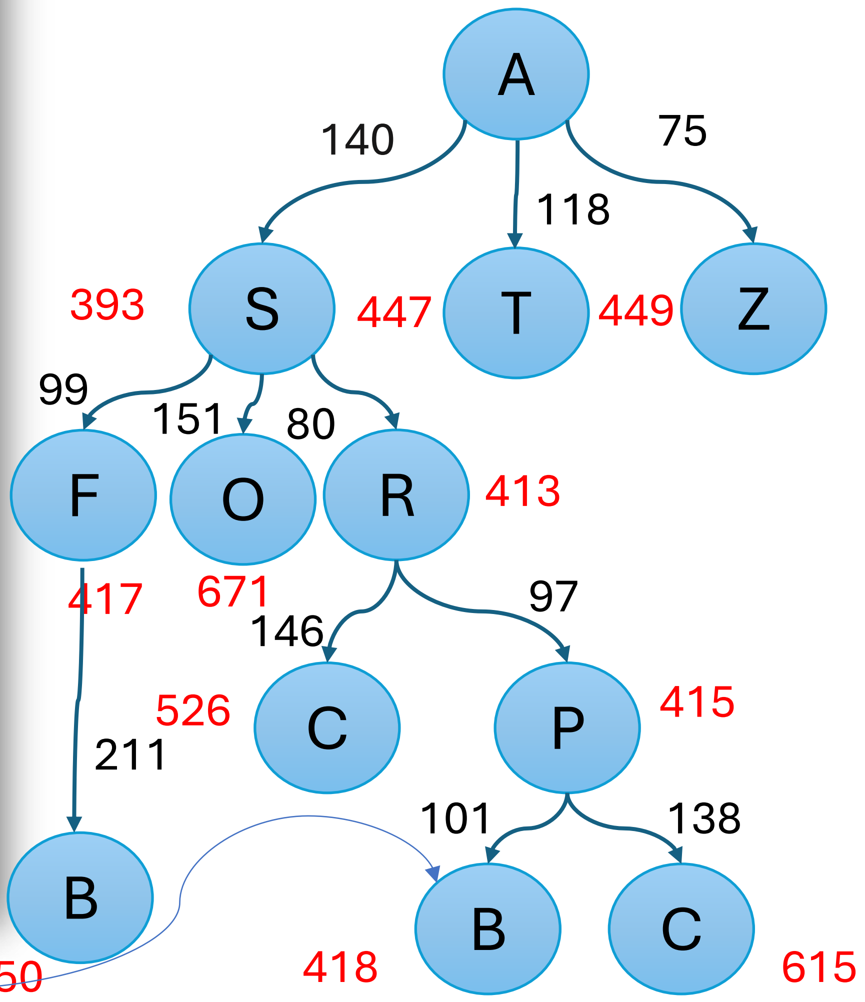
1. Given the **Arad/Bucharest** route finding problem and the heuristics table below, show the list n of visited nodes using the following search techniques:

Order : A ,S, R,P,F,B
Path : A,S,R,P,B

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374



A*



Now the Goal Node is the one with the least Cost in the Whole Tree Stop

Using the A* algorithm work out a route from town A to town M. Use the following cost functions.

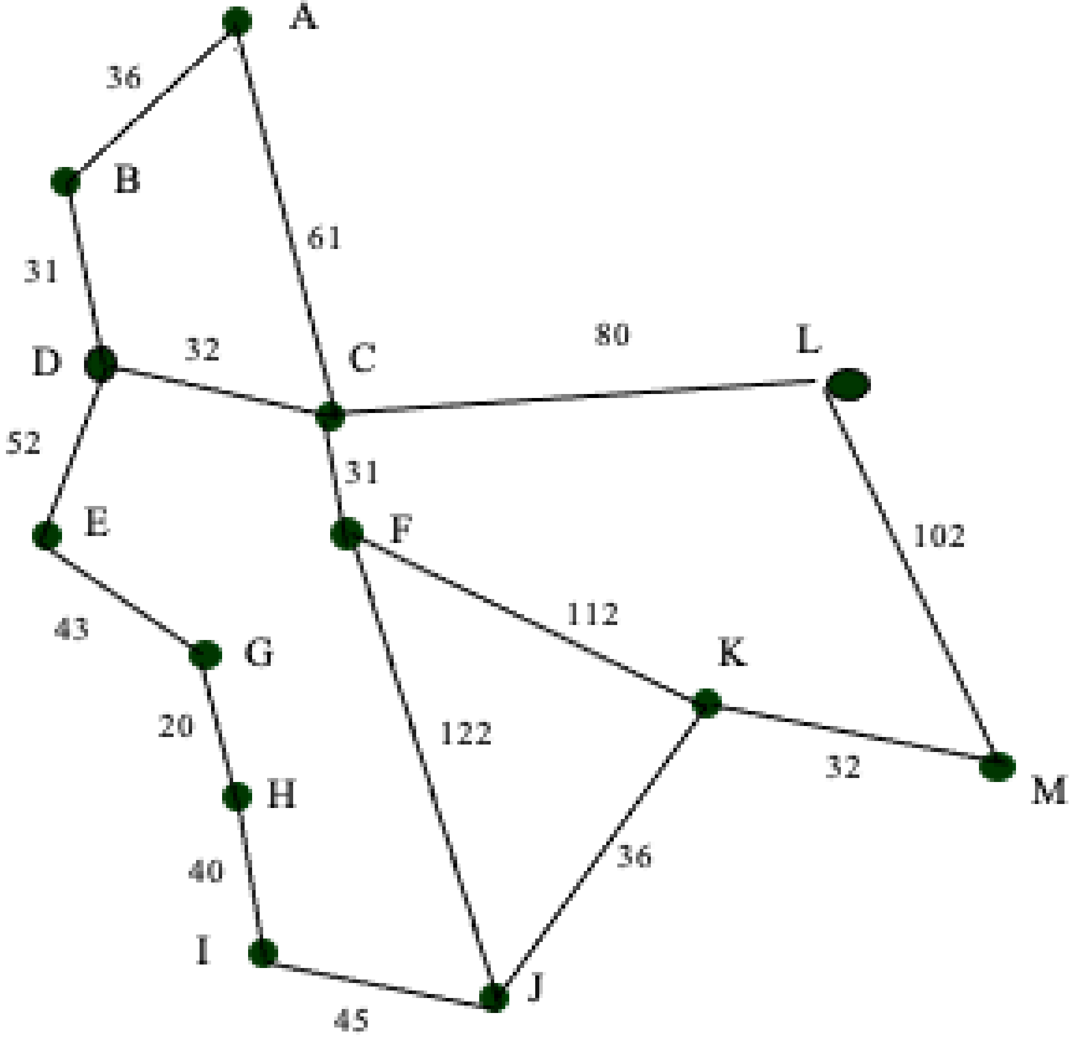
Straight Line Distance to M

A	223
B	222
C	166
D	192

E	165
F	136
G	122
H	111

I	100
J	60
K	32
L	102

M	0



A*

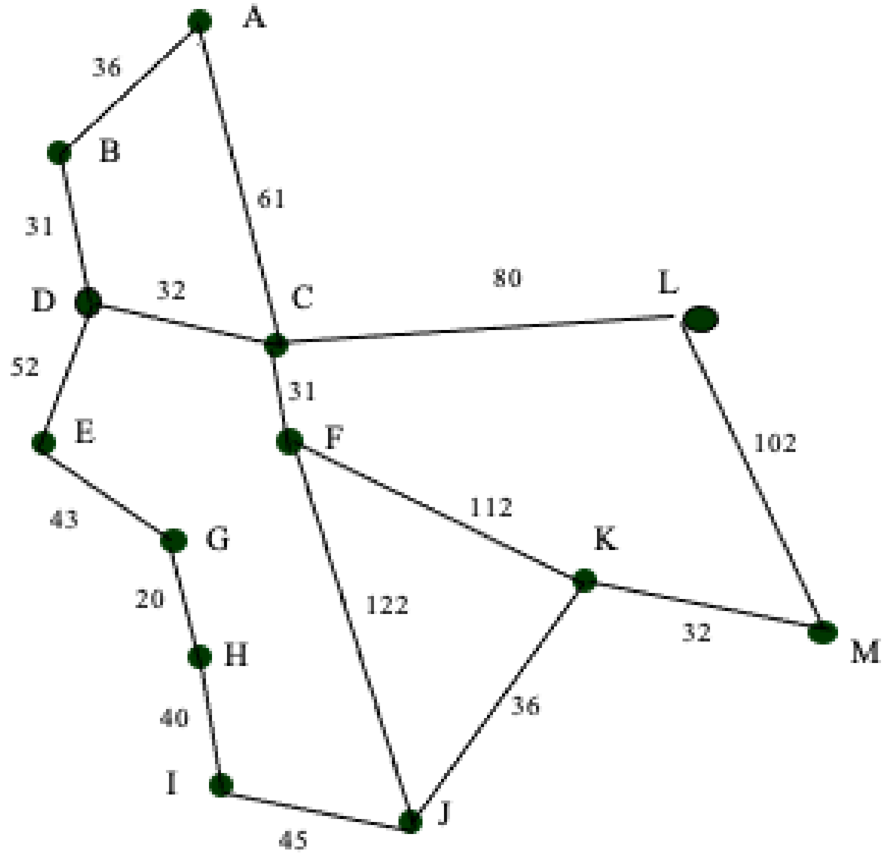
Straight Line Distance to M

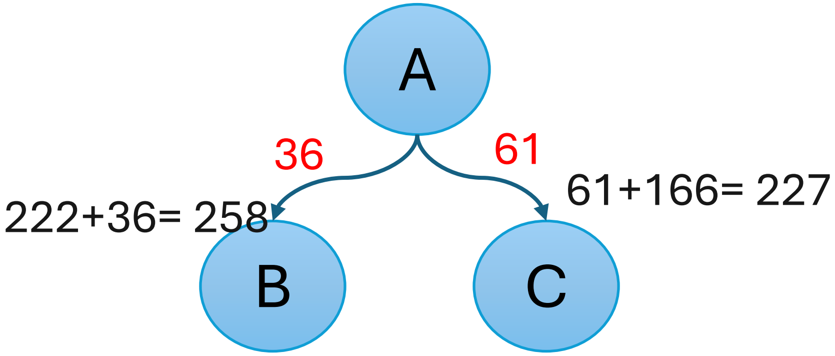
A	223
B	222
C	166
D	192

E	165
F	136
G	122
H	111

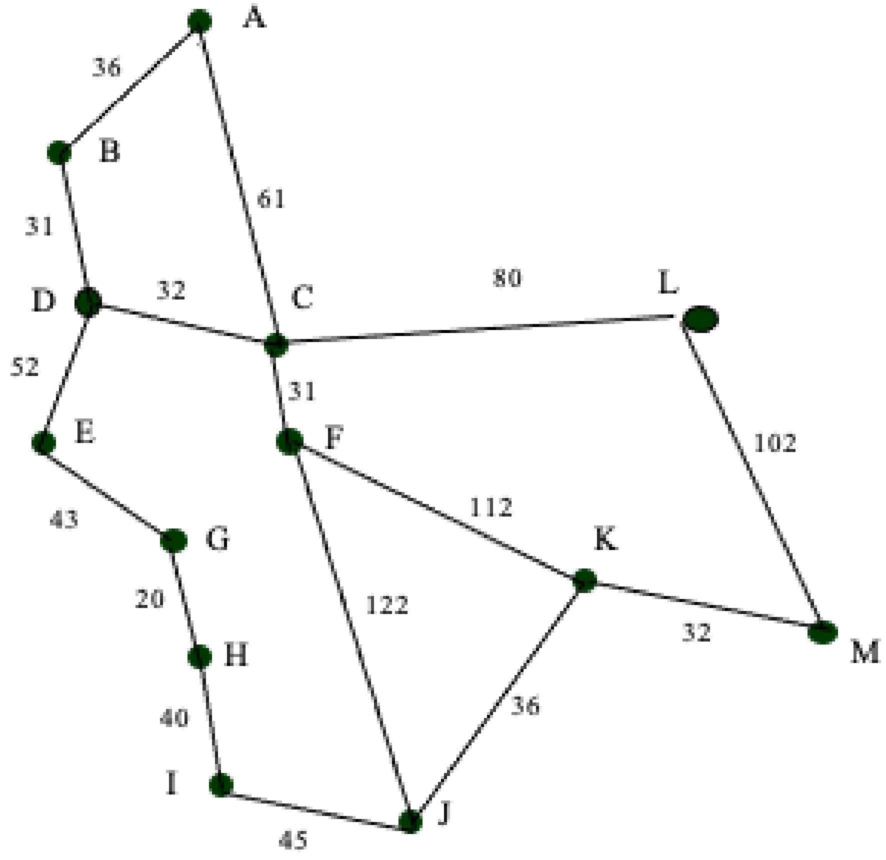
I	100
J	60
K	32
L	102

M	0





A*



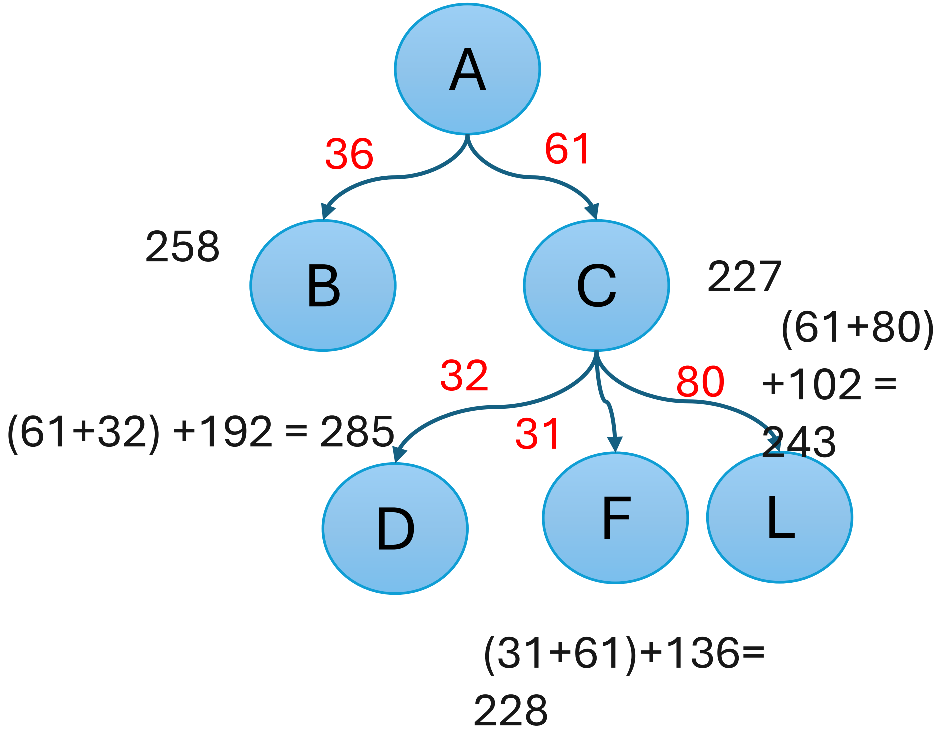
Straight Line Distance to M

A	223
B	222
C	166
D	192

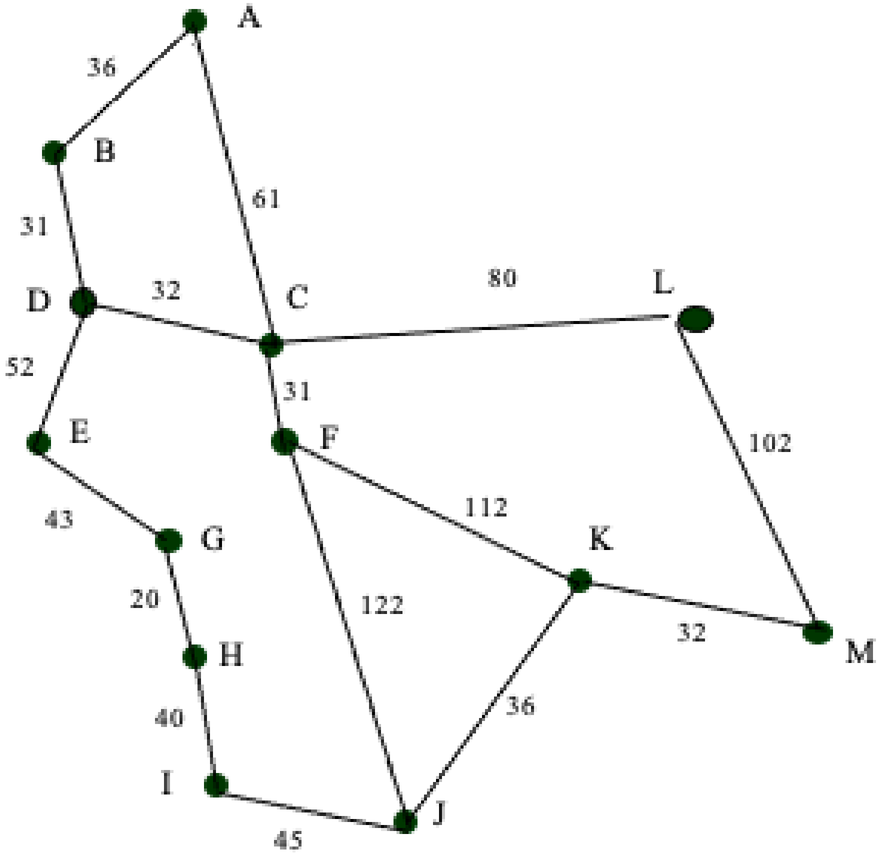
E	165
F	136
G	122
H	111

I	100
J	60
K	32
L	102

M	0



A*



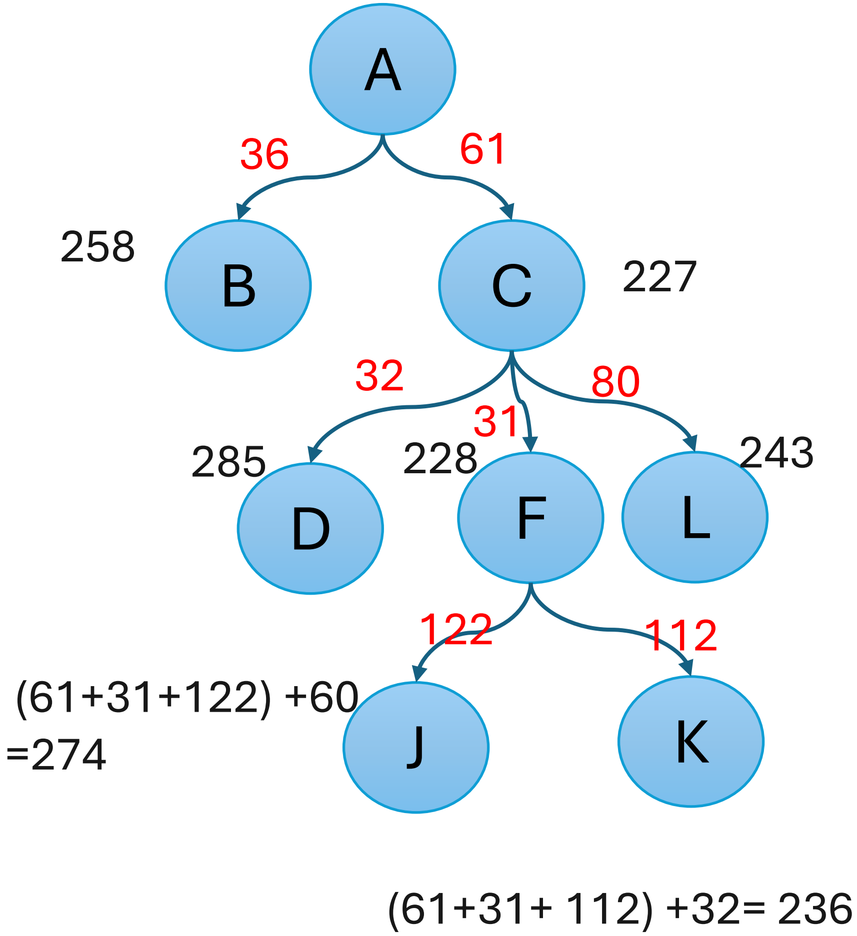
Straight Line Distance to M

A	223
B	222
C	166
D	192

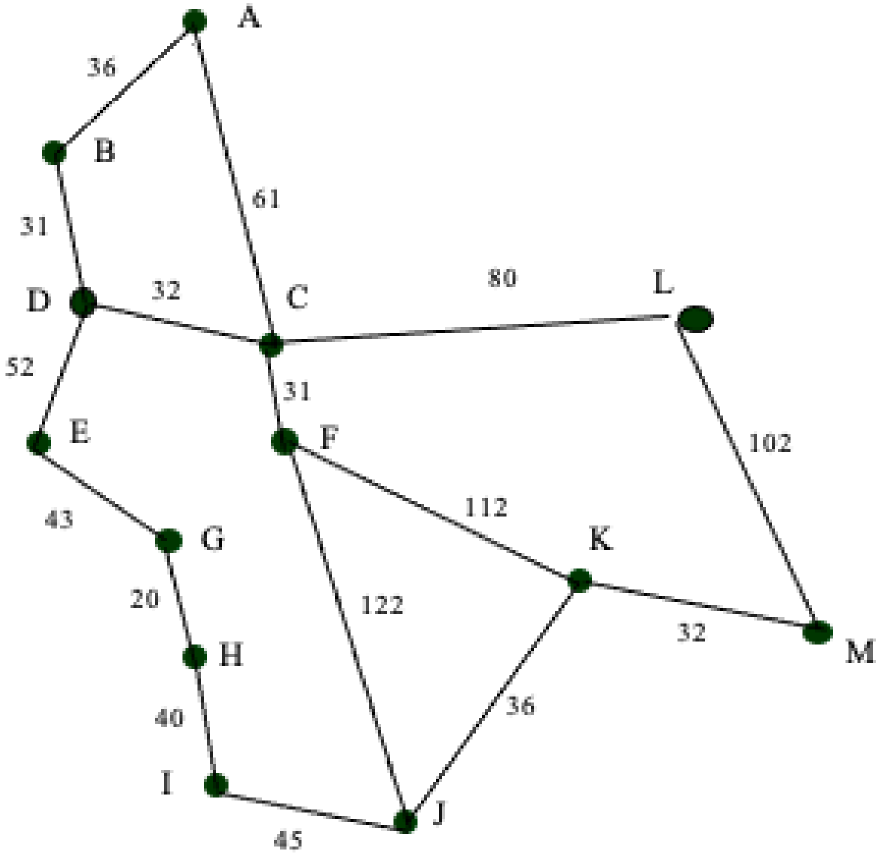
E	165
F	136
G	122
H	111

I	100
J	60
K	32
L	102

M	0

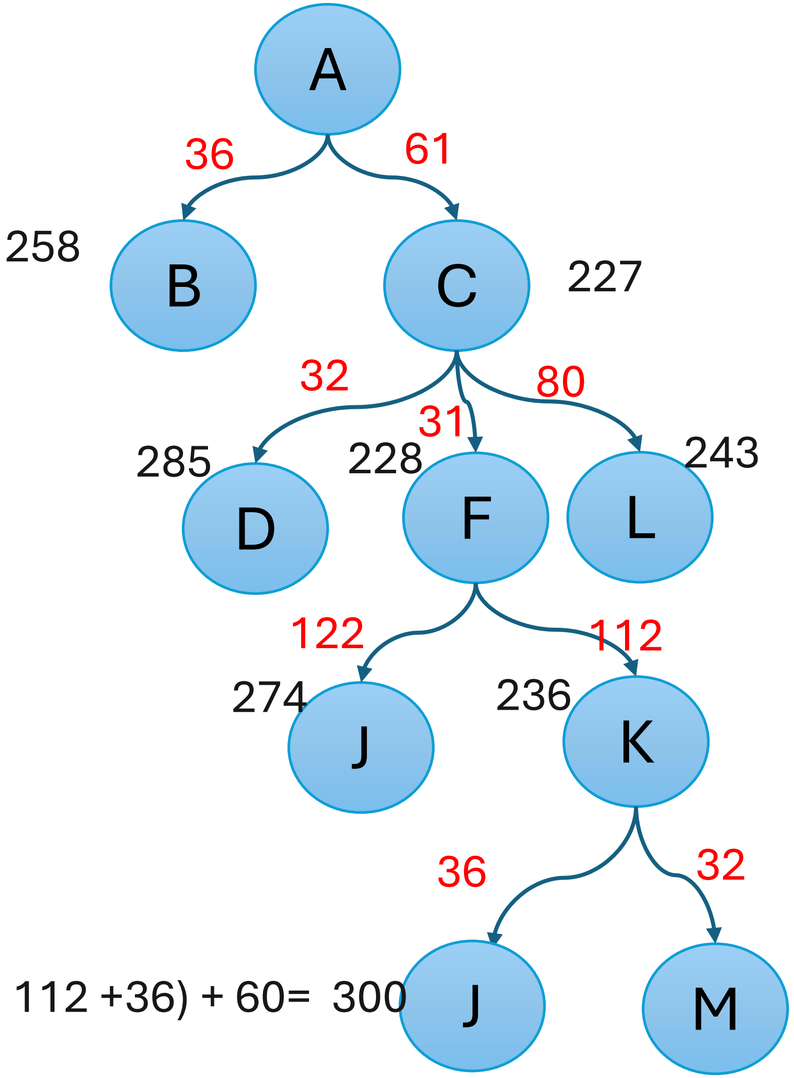


A*



Straight Line Distance to M

A	223
B	222
C	166
D	192
E	165
F	136
G	122
H	111
I	100
J	60
K	32
L	102
M	0



(61+31+ 112 +36) + 60= 300

(61+31+ 112 +32) + 0= 236

A*

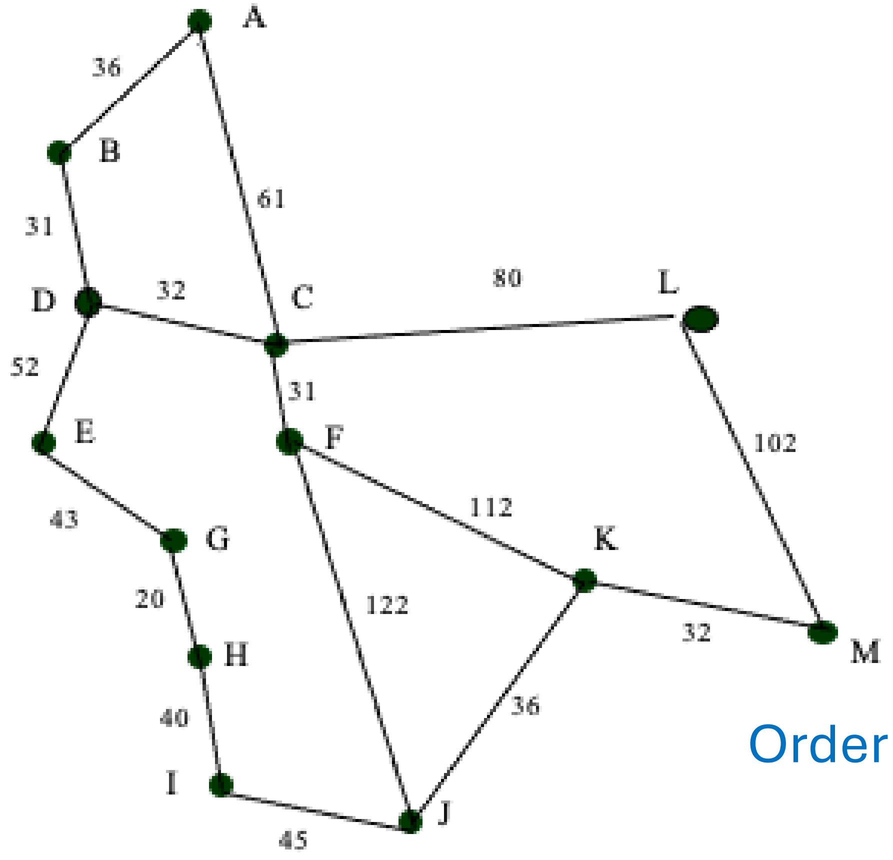
Straight Line Distance to M

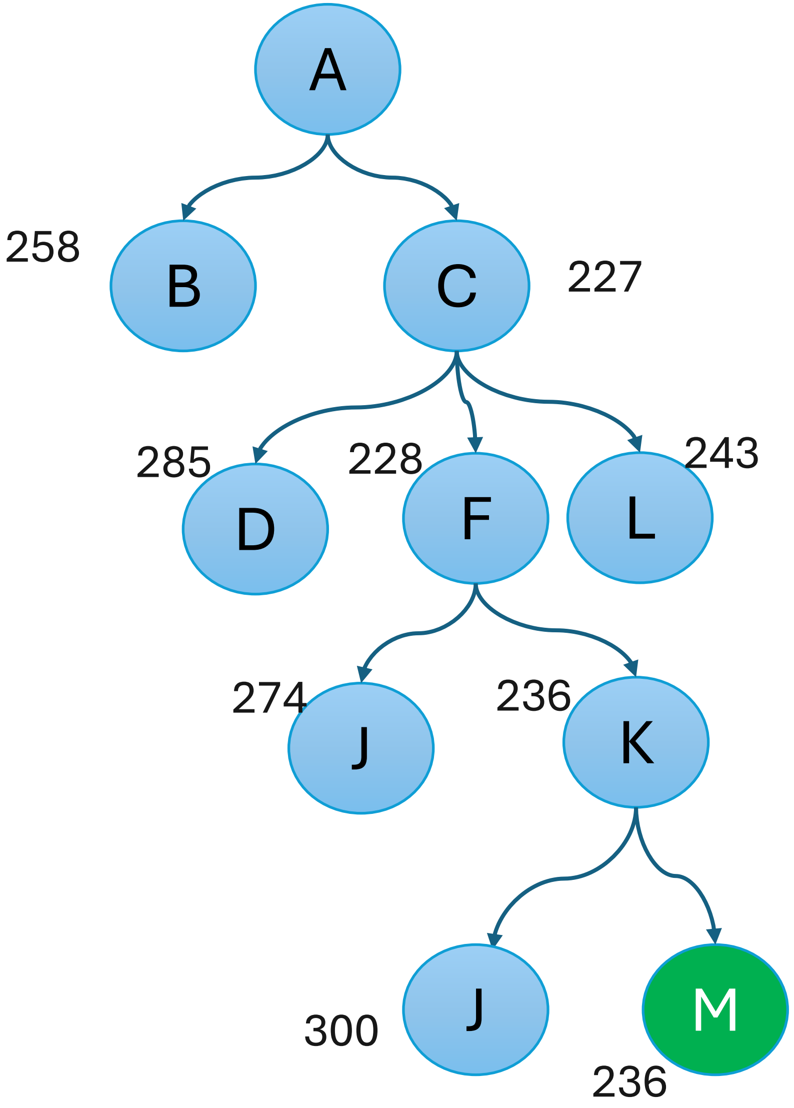
A	223
B	222
C	166
D	192

E	165
F	136
G	122
H	111

I	100
J	60
K	32
L	102

M	0





Order: A, C, F, K, M

Path: A, C, F, K, M